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Learn Beyond

AIRWAY RESISTANCE MEASUREMENT

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AIRWAY RESISTANCE MEASUREMENT

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- Airway resistance can be determined by simultaneously measuring the intra-alveolar pressure and airflow in the body plethysmograph and by dividing the difference between the intra-alveolar pressure and the atmospheric pressure by the flow.

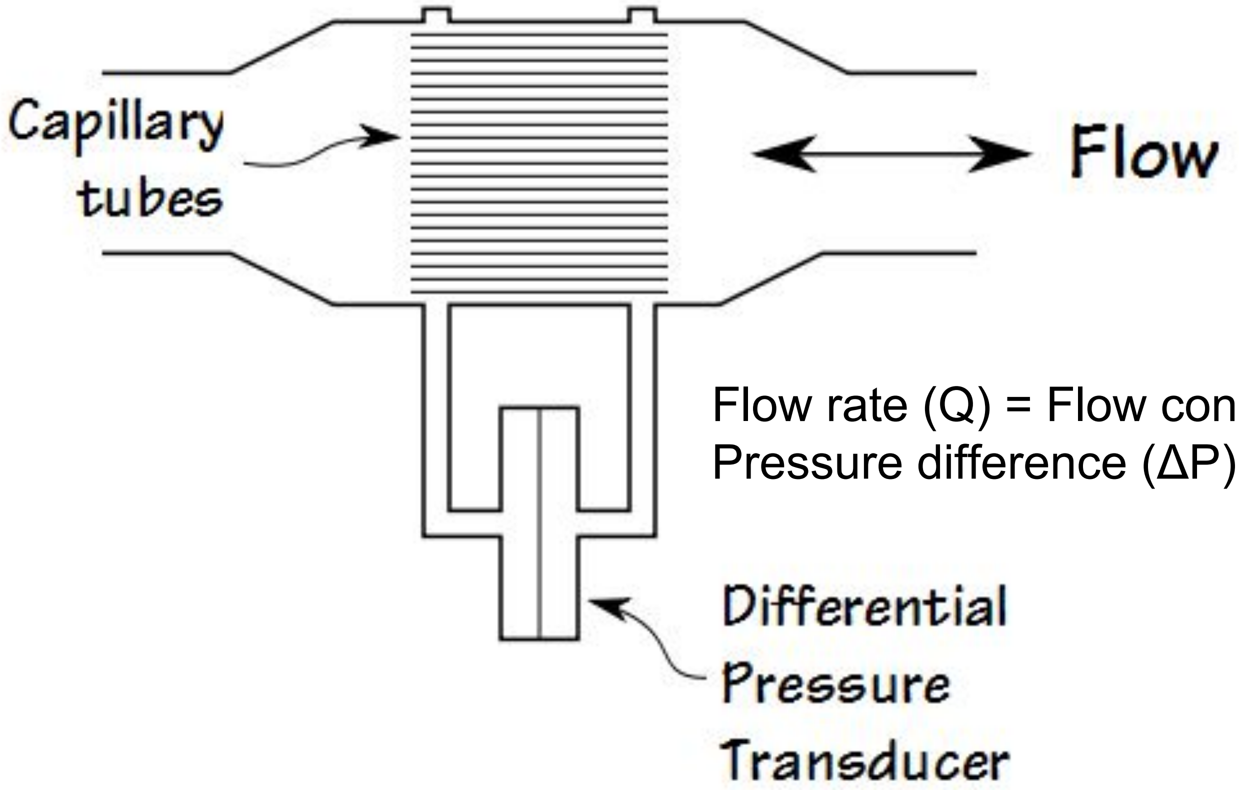
$$R = \frac{P_{ia} - P_A}{f}$$

- where R = airway resistance, P_{ia} = intra-alveolar pressure, P_A = atmospheric pressure and f = airflow.

PNEUMOTACHOMETER

- A variety of instruments can be used to measure airflow.
- One of the most widely used is the pneumotachometer.
- This device utilizes the principle that air flowing through an orifice produces a pressure difference across the orifice that is a function of the velocity of the air.
- In the more common pneumotachometer, the orifice consists of a set of capillaries or a metal screen.
- Since the cross section of the orifice is fixed, the pressure difference can be calibrated to represent flow.
- Two pressure transducers or a differential pressure transducer can be used to measure the pressure difference.

PNEUMOTACHOMETER



Flow rate (Q) = Flow constant (K) X Pressure difference (ΔP)

- Another method of measuring airflow is a transducer in which a heated wire is cooled by the flow of air, and the resistance change due to the cooling is measured as representative of airflow.
- Because the cooling effect is the same regardless of the direction of airflow, this transducer is insensitive to direction, whereas the pneumotachograph described above indicates not only the amount of flow but also the direction.

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- Ultrasonic airflow-measuring devices utilizing the Doppler effect have been developed.
- Since flow is the first derivative or rate of change of volume, some volume-measuring devices also produce a measurement of flow.
- Also in use is a small breath-driven turbine which operates a miniature electrical generator that produces an output voltage proportional to the air velocity.

- For measurement of intrathoracic pressures, a balloon is placed in the patient's esophagus, which is within the thoracic cage.
- Since the balloon is exposed to the intrathoracic pressure, its pressure, measured with respect to mouth pressure by using some form of differential pressure transducer, represents the difference between pressures.

LUNG VOLUMES AND CAPACITIES

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LUNG VOLUME

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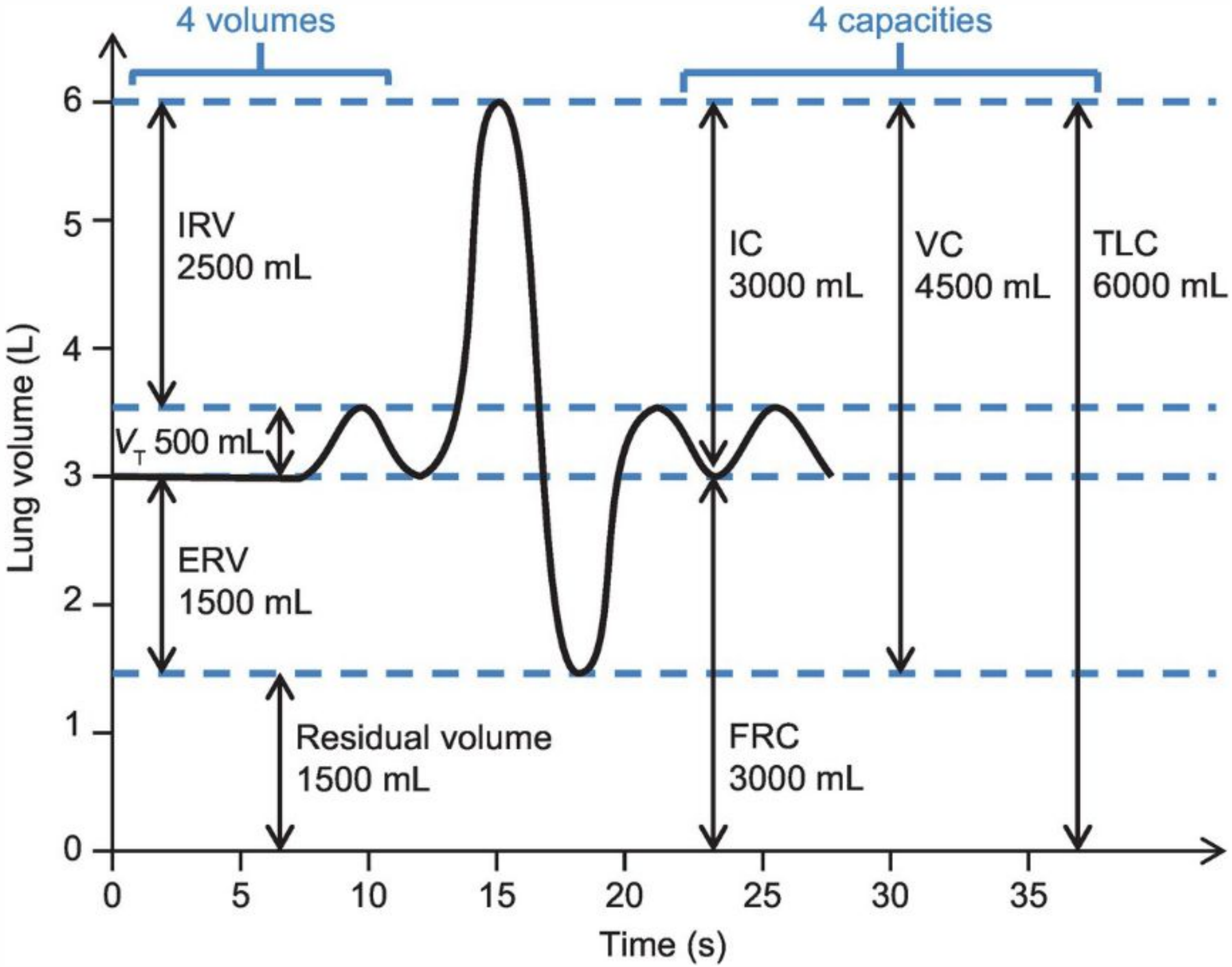
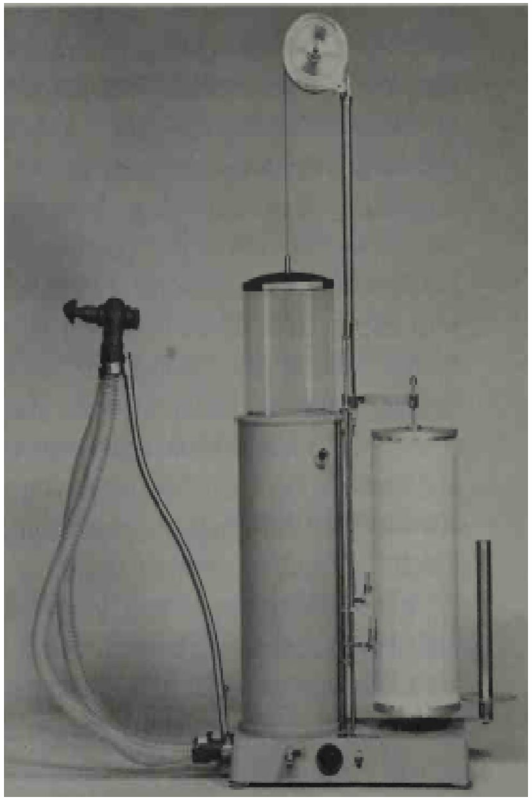
- The largest volume to which the subject's lungs can be voluntarily expanded is defined as the total lung capacity (TLC).
- The smallest volume to which the subject can slowly deflate his or her lungs is the residual volume (RV).
- The volume of the lungs at the end of a quiet expiration when the respiratory muscles are relaxed is the functional residual capacity (FRC).
- Vital capacity defines the maximal change in volume the lungs can undergo during respiration.

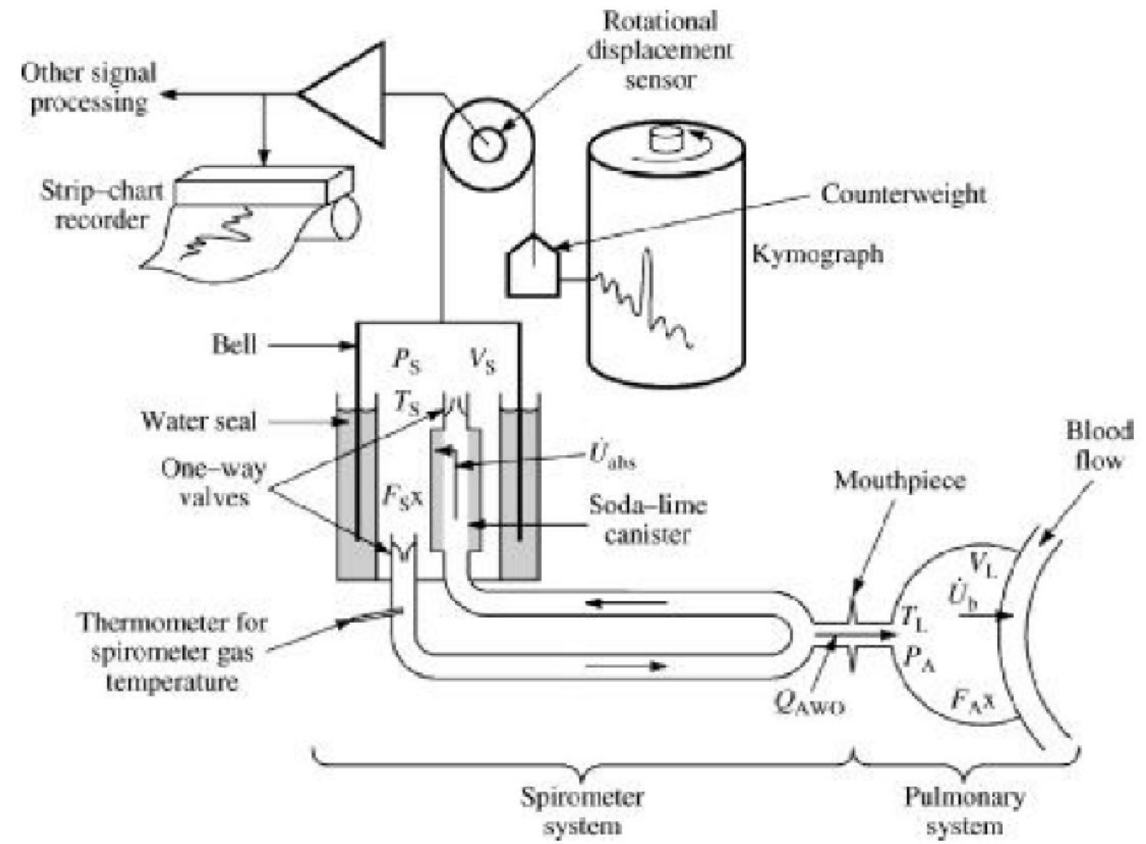

4 volumes
4 capacities
IRV
2500 mL
V_T 500 mL
ERV
1500 mL
Residual volume
1500 mL
IC
3000 mL
VC
4500 mL
TLC
6000 mL
FRC
3000 mL
Lung volume (L)
Time (s)
0
1
2
3
4
5
6
0
5
10
15
20
25
30
35

SPIROMETER

- The standard spirometer consists of a movable bell inverted over a chamber of water.
- Inside the bell, above the water line, is the gas that is to be breathed.
- The bell is counterbalanced by a weight to maintain the gas inside at atmospheric pressure so that its height above the water is proportional to the amount of gas in the bell.
- A breathing tube connects the mouth of the patient with the gas under the bell.
- Thus, as the patient breathes into the tube, the bell moves up and down with each inspiration and expiration in proportion to the amount of air breathed in or out.

SPIROMETER





Rotational displacement sensor
Other signal processing
Strip-chart recorder
Counterweight
Kymograph
Bell
P_S
V_S
Water seal
T_S
$\dot{U}_{abs}$
One-way valves
$F_S x$
Soda-lime canister
Mouthpiece
Blood flow
V_L
$\dot{U}_b$
T_L
P_A
Thermometer for spirometer gas temperature
Q_{AWO}
$F_A x$
Spirometer system
Pulmonary system

SPIROMETER

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- Attached to the bell or the counterbalancing mechanism is a pen that writes on an adjacent drum recorder, called a kymograph.
- As the kymograph rotates, the pen traces the breathing pattern of the patient.
- Various bell volumes are available, but 9 and 13.5 liters are most common.
- A well-designed spirometer offers little resistance to airflow, and the bell has little inertia.
- Various paper speeds are available for the kymograph, with 32, 160, 300, and 1920 mm/min most common.

SPIROMETER

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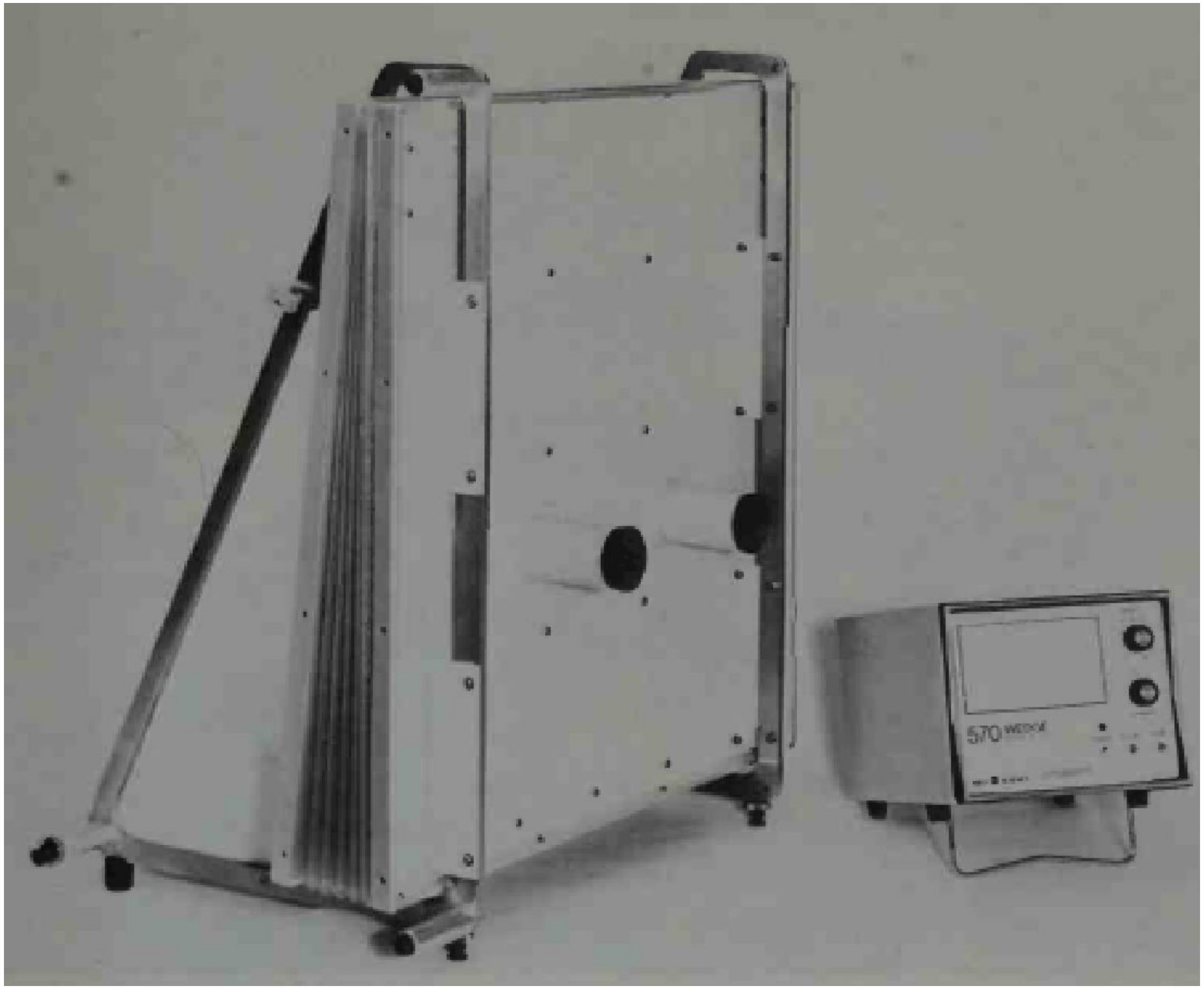
- Its 9-liter capacity is often considered adequate for recording the largest vital capacities.
- Many physicians prefer the larger size (13.5 liters) because of the extra capacity.
- This instrument is equally suitable for clinical spirography, for cardiopulmonary function testing, and for metabolism determinations.
- The instrument directly records basal minute volume, exercise ventilation, or maximum breathing capacity.
- The ventilation equivalent for oxygen may be calculated directly from the spirogram slope lines for ventilation and oxygen uptake.

WATERLESS SPIROMETER

- Also called Wedge spirometer.
- In this instrument the air to be breathed is held in a chamber enclosed by two parallel metal pans hinged to each other along one edge.
- The space between the two pans is enclosed by a flexible bellows to form the chamber.
- One of the pans, which contains an inlet tube, is fixed to a stand and the other swings freely with respect to it.
- As air is introduced into the chamber or withdrawn from it, the moving pan changes its position to compensate for the volume changes.
- The instrument provides electrical outputs proportional to both volume and airflow, from which the required determinations can be obtained.

WATERLESS SPIROMETER



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WATERLESS SPIROMETER

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- In a similar type of waterless spirometer, the volume of the chamber is varied by means of a lightweight piston that moves freely in a cylinder as air is withdrawn and replaced in breathing.
- A Silastic rubber seal between the piston and the cylinder wall keeps the chamber airtight.
- Instruments of this type have characteristics similar to those of the wedge spirometer.

ELECTRONIC SPIROMETER

- Electronic spirometers, measures airflow and, by use of electronic circuitry, calculates the various volumes and capacities.
- This instrument provides both a graphic output similar to that of a standard spirometer and a digital readout of the desired parameters.
- Various types of airflow transducers are used, utilizing such devices as small breath-driven turbines and heated wires that are cooled by the breath.

ELECTRONIC SPIROMETER

BRONCHOSPIROMETER

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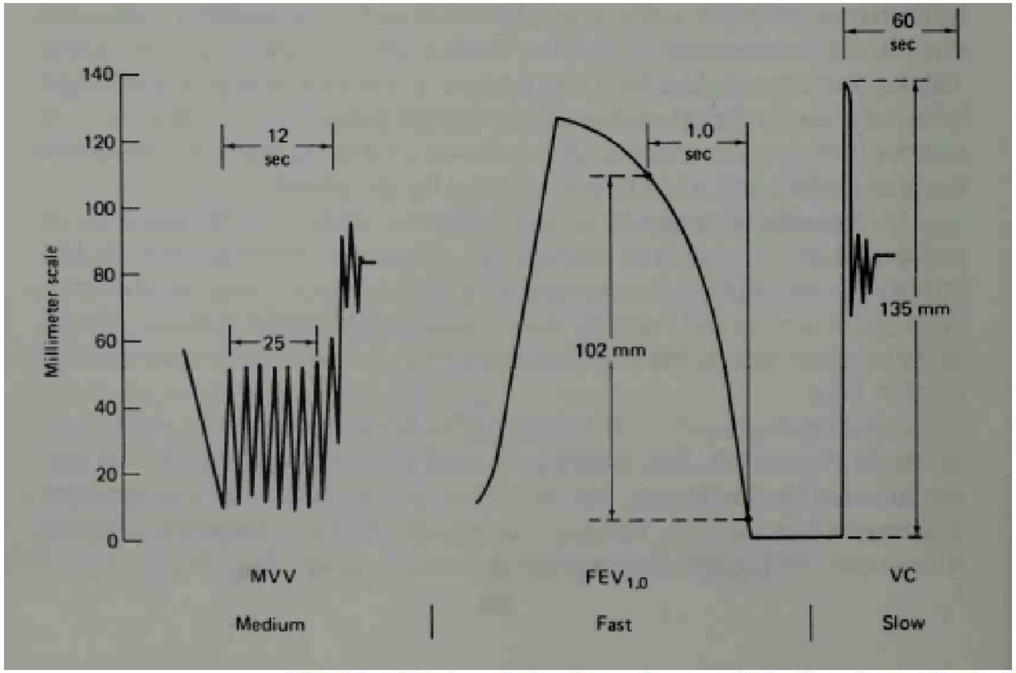
- A bronchospirometer is a dual spirometer that measures the volumes and capacities of each lung individually.
- The air-input device is a double lumen tube that divides for entry into the airway to each lung and thus provides isolation for differential measurement.
- The main function of the bronchospirometer is the preoperative evaluation of oxygen consumption of each lung.

SPIROGRAM

- The output of a spirometer is the spirogram.
- The recording is read from right to left.
- Inspiration moves the pen toward the bottom of the chart and expiration toward the top.
- Some spirometers, provide spirograms with inspiration toward the top.



LIMITATIONS OF SPIROMETER

- All the lung volumes and capacities can be determined except those that require measurement of the air still remaining in the lungs and airways after maximum expiration.
- These parameters, which include the residual volume, FRC, and total lung capacity, can be measured through the use of foreign gas mixtures.
- A gas analyzer is required for these tests.

INHALATORS, HUMIDIFIERS AND NEBULIZERS

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INHALATORS

- Inhalator is a device used to supply oxygen or some other therapeutic gas to a patient who is able to breathe spontaneously without assistance.
- As a rule, inhalators are used when a concentration of oxygen higher than that of air is required.
- The inhalator consists of a source of the therapeutic gas, equipment for reducing the pressure and controlling the flow of the gas, and a device for administering the gas.
- Devices for administering oxygen to patients include nasal cannulae and catheters, face masks that cover the nose and mouth, and, in certain settings, such as pediatrics, oxygen tents.
- The oxygen concentration presented to the patient is controlled by adjusting the flow of gas into the mask.

HUMIDIFERS

- In order to prevent damage to the patient's lungs, the air or oxygen applied during respiratory therapy must be humidified.
- Thus, virtually all inhalators, ventilators, and respirators include equipment to humidify the air, either by heat vaporization (steam) or by bubbling an air stream through a jar of water.

NEBULIZER

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- When therapy requires that water or some type of medication be suspended in the inspired air as an aerosol, a device called a nebulizer is used.
- In a nebulizer the water or medication is picked up by a high-velocity jet of oxygen (or some other gas) and thrown against one or more baffles or other surfaces to break the substance into controllable-sized droplets or particles, which are then applied to the patient via a respirator.

ULTRASONIC NEBULIZER

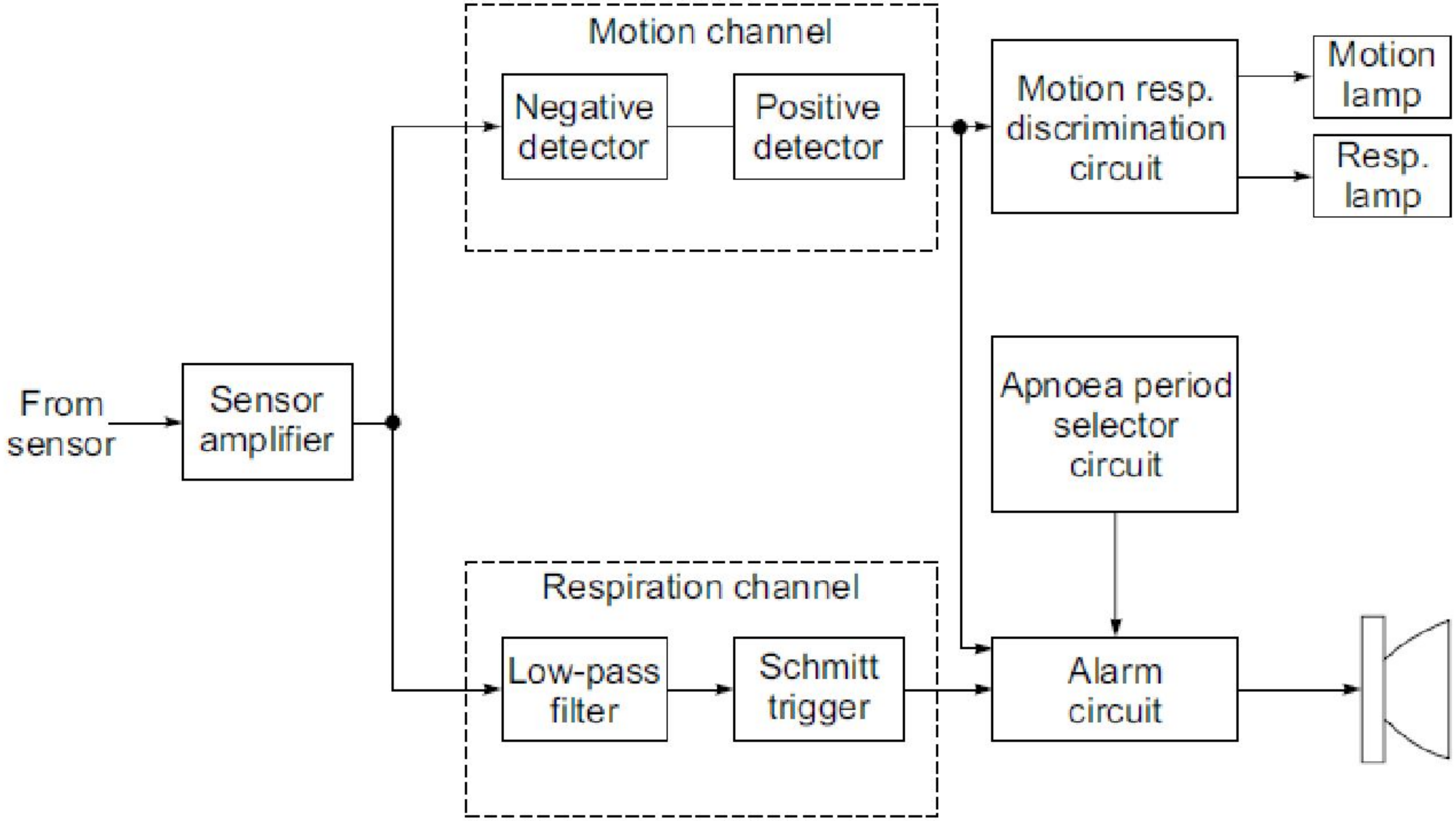
- A more effective type of nebulizer is the ultrasonic nebulizer.
- This electronic device produces high-intensity sound energy well above the audible range.
- When applied to water or medication, the ultrasonic energy vibrates the substance with such intensity that a high volume of minute particles is produced.
- Such equipment usually consists of two parts, a generator that produces a radiofrequency current to drive the ultrasonic transducer, and the nebulizer itself, in which the transducer generates the ultrasound energy and applies it to the water or medication.

APNOEA MONITOR

- Apnoea is the cessation of breathing which may precede the arrest of the heart and circulation in several clinical situations such as head injury, drug overdose, anaesthetic complications and obstructive respiratory diseases.
- Apnoea may also occur in premature babies during the first weeks of life because of their immature nervous system.
- If apnoea persists for a prolonged period, brain function can be severely damaged.
- Therefore, apnoeic patients require close and constant observation of their respiratory activity.
- Apnoea monitors are particularly useful for monitoring the respiratory activity of premature infants.

- The apnoea monitors are basically motion detectors and are thus subject to other motion artefacts also which could give false readings.
- The instruments must, therefore, provide means of elimination of these error sources.
- It basically consists of an input amplifier circuit, motion and respiration channels, a motion/respiration discrimination circuit, and an alarm circuit.
- The input circuit consists of a high input impedance amplifier which couples the input signal from the sensor pad to the logic circuits.
- The sensor may be a strain gauge transducer embedded in the mattress.

- The output of the amplifier is adjusted to zero volts with offset adjustment provided in the amplifier.
- The amplified signal goes to motion and respiration channels connected in parallel.
- The motion channel discriminates between motion and respiration as a function of frequency.
- In the case of motion signals, high level signals above a fixed threshold are detected from the sensor.
- In the respiration channel, a low-pass filter is trigger circuit to pulse at the respiration rate.
- Higher frequency signals, above 1.5 Hz (motion), cause the output of the trigger to go positive.
- Absence of the signal (apnoea) causes the output of the Schmidt trigger to go negative.


From sensor
Sensor amplifier
Motion channel
Negative detector
Positive detector
Motion resp. discrimination circuit
Motion lamp
Resp. lamp
Apnoea period selector circuit
Respiration channel
Low-pass filter
Schmitt trigger
Alarm circuit

MEASUREMENT OF RESIDUAL VOLUME

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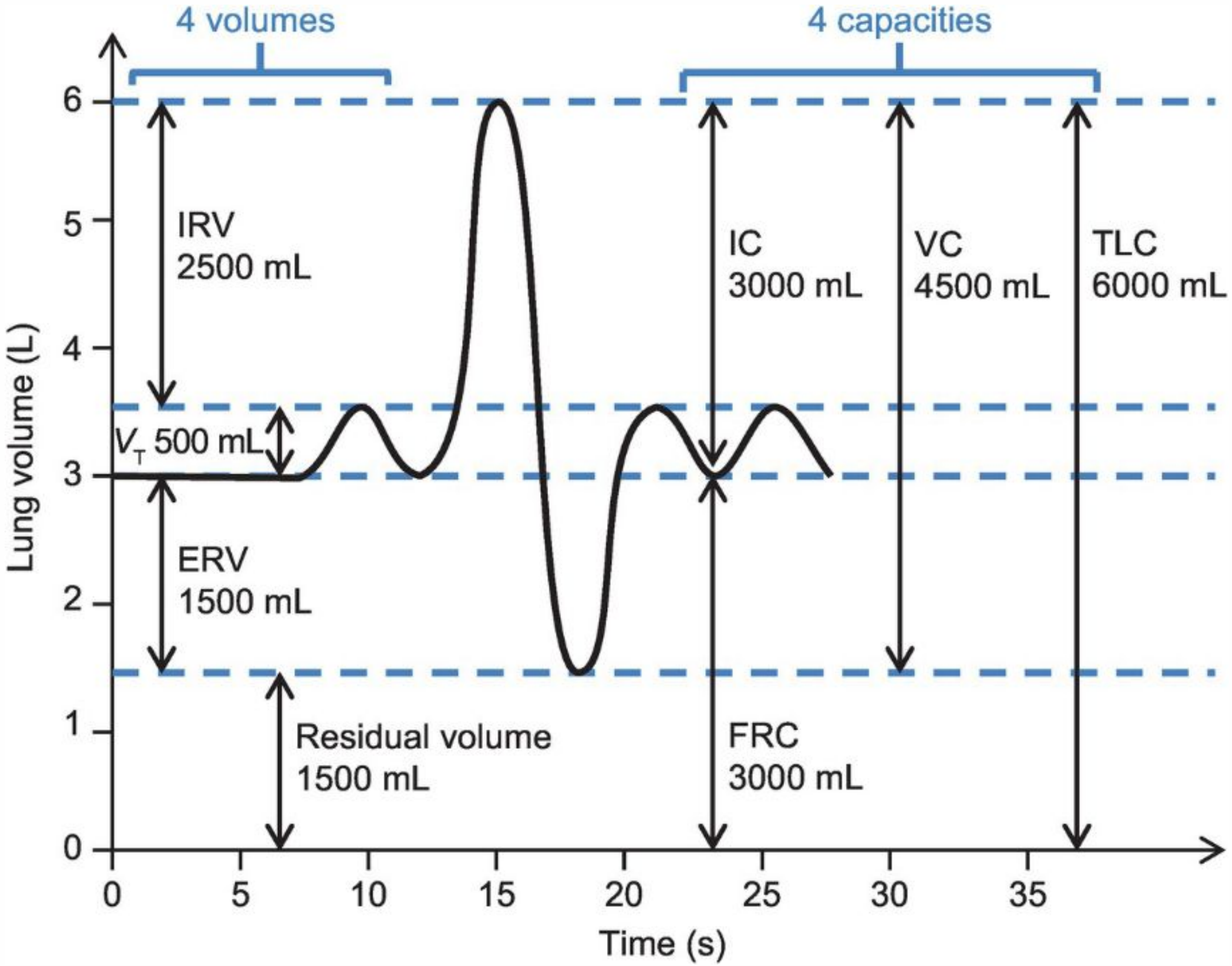
LIMITATIONS OF SPIROMETER

- All the lung volumes and capacities can be determined except those that require measurement of the air still remaining in the lungs and airways after maximum expiration.
- These parameters, which include the residual volume, FRC, and total lung capacity, can be measured through the use of foreign gas mixtures.
- A gas analyzer is required for these tests.


4 volumes
4 capacities
Lung volume (L)
6
5
4
3
2
1
0
IRV
2500 mL
V_T 500 mL
ERV
1500 mL
Residual volume
1500 mL
IC
3000 mL
VC
4500 mL
TLC
6000 mL
FRC
3000 mL
0
5
10
15
20
25
30
35
Time (s)

CLOSED CIRCUIT TECHNIQUE

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- The closed-circuit technique involves rebreathing from a spirometer charged with a known volume and concentration of a marker gas, such as hydrogren or helium.
- Helium is usually used.
- After several minutes of breathing, complete mixing of the spirometer and pulmonary gases is assumed, and the residual volume is calculated by a simple proportion of gas volumes and concentrations.

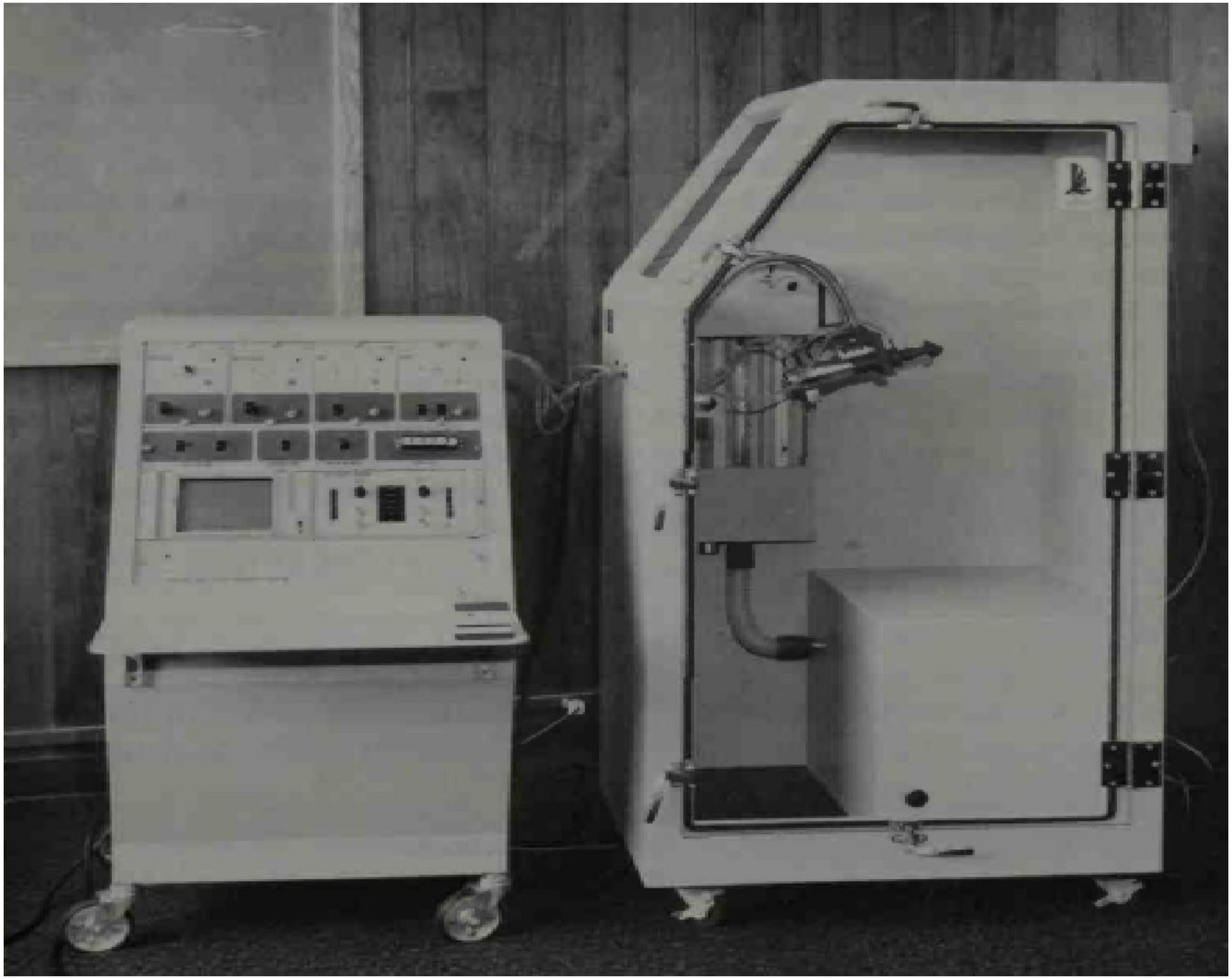
OPEN CIRCUIT / NITROGEN WASH OUT METHOD

- The open-circuit or nitrogen washout method involves the inspiration of pure oxygen and expiration into an oxygen-purged spirometer.
- If the patient has been breathing air, the gas remaining in his lungs is 78 percent nitrogen.
- As he begins to breathe the pure oxygen, it will mix with the gas still in his lungs, and a certain amount of nitrogen will "wash out" with each breath.
- By measuring the amount of nitrogen in each expired breath, a washout curve is obtained from which the volume of air initially in the lungs can readily be calculated.

PLETHYSMOGRAPH

- The functional residual capacity (FRC) can be measured by using a body plethysmograph.
  - This instrument, is an airtight box in which the patient is seated.
  - Utilizing Boyle's law (at constant temperature, the volume of gas varies inversely with the pressure), the ratio of the change in lung volume to change in mouth pressure is used to determine the thoracic gas volume.
  - The patient breathes air from within the box through a tube containing an airflow transducer and a shutter to close off the tube for certain portions of the test.

PLETHYSMOGRAPH



- Pressure transducers measure the air pressure in the breathing tube on the patient's side of the shutter and inside the box.
- The amount of air in the box, including that in the patient's lungs, remains constant, since there is no way for air to enter or escape.
- However, when the patient compresses the air in his lungs during expiration, his total body volume is reduced, thus reducing the pressure in the box.
- Conversely, when the patient inhales by reducing the pressure in his thoracic region, his body volume increases and increases the box pressure.

- The FRC is measured with the shutter in the breathing tube closed.
- With no air allowed to flow, the mouth pressure can be assumed to equal the alveolar pressure.
- The patient is instructed to pant at a slow rate against the closed shutter.
- As he does so, he alternately expands and compresses the air in his lungs.
- By measuring the changes in mouth pressure and corresponding changes in intrathoracic volume (Equal and opposite to changes in box volume outside the patient), it is possible to calculate the intrathoracic volume.
- If the test is performed at the end expiratory level, the intrathoracic volume is equal to the FRC.

INTRA-ALVEOLAR AND THORACIC PRESSURE MEASUREMENTS

- The body plethysmograph can also be used to measure intra-alveolar and intrathoracic pressures.
- These measurements are important in the determination of both compliance and airway resistance, since inaccessibility of these chambers makes direct measurement impossible.
- For measurement of intraalveolar pressures, the shutter in the breathing tube is opened to allow the patient to freely breathe air from within the closed box.

- Since the patient and the box form a closed system containing a fixed amount of gas, pressure and volume variations in the box are the inverse of the pressure variations in the lungs as the gas within the lungs expands and is compressed due to the positive and negative pressures in the lungs.
- For calibration, the patient's breathing tube is blocked for a few seconds, during which the patient is asked to the “pant” while mouth pressure is measured.
- Since mouth pressure and lung pressure are the same when there is no airflow, these data can be used in calibration of the measurement.

- For measurement of intrathoracic pressures, a balloon is placed in the patient's esophagus, which is within the thoracic cage.
- Since the balloon is exposed to the intrathoracic pressure, its pressure, measured with respect to mouth pressure by using some form of differential pressure transducer, represents the difference between pressures.

VENTILATORS

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VENTILATORS

- Ventilator – improve ventilation of the lungs and to supply humidity or aerosol medications to the pulmonary tree.
- Most ventilators in clinical settings use positive pressure during inhalation to inflate the lungs with various gases or mixtures of gases (air, oxygen, carbon dioxide, helium, etc.).
- Expiration is usually passive, although under certain conditions pressure may be applied during the expiratory phase as well, in order to improve arterial oxygen tension.

MODES OF VENTILATION

- Three modes of ventilation:
  - Assist mode
  - Control
  - Assist – control

ASSIST MODE

- In the assist mode inspiration is triggered by the patient.
- A pressure sensor responds to the slight negative pressure that occurs each time the patient attempts to inhale and triggers the apparatus to begin inflating the lungs.
- Thus, the respirator helps the patient inspire when he wants to breathe.
- The assist mode is used for patients who are able to control their breathing but are unable to inhale a sufficient amount of air without assistance or for whom breathing requires too much effort.

CONTROL MODE

- In the control mode breathing is controlled by a timer set to provide the desired respiration rate.
- Controlled ventilation is required for patients who are unable to breathe on their own.
- In this mode the respirator has complete control over the patient's respiration and does not respond to any respiratory effort on the part of the patient.

ASSIST CONTROL MODE

- In the assist-control mode the apparatus is normally triggered by the patient's attempts to breathe, as in the assist mode.
- If the patient fails to breathe within a predetermined time, a timer automatically triggers the device to inflate the lungs.
- Thus, the patient controls his own breathing as long as he can, but if he should fail to do so, the machine is able to take over for him.
- This mode is most frequently used in critical care settings.

- For measurement of intrathoracic pressures, a balloon is placed in the patient's esophagus, which is within the thoracic cage.
- Since the balloon is exposed to the intrathoracic pressure, its pressure, measured with respect to mouth pressure by using some form of differential pressure transducer, represents the difference between pressures.

TYPES OF VENTILATORS

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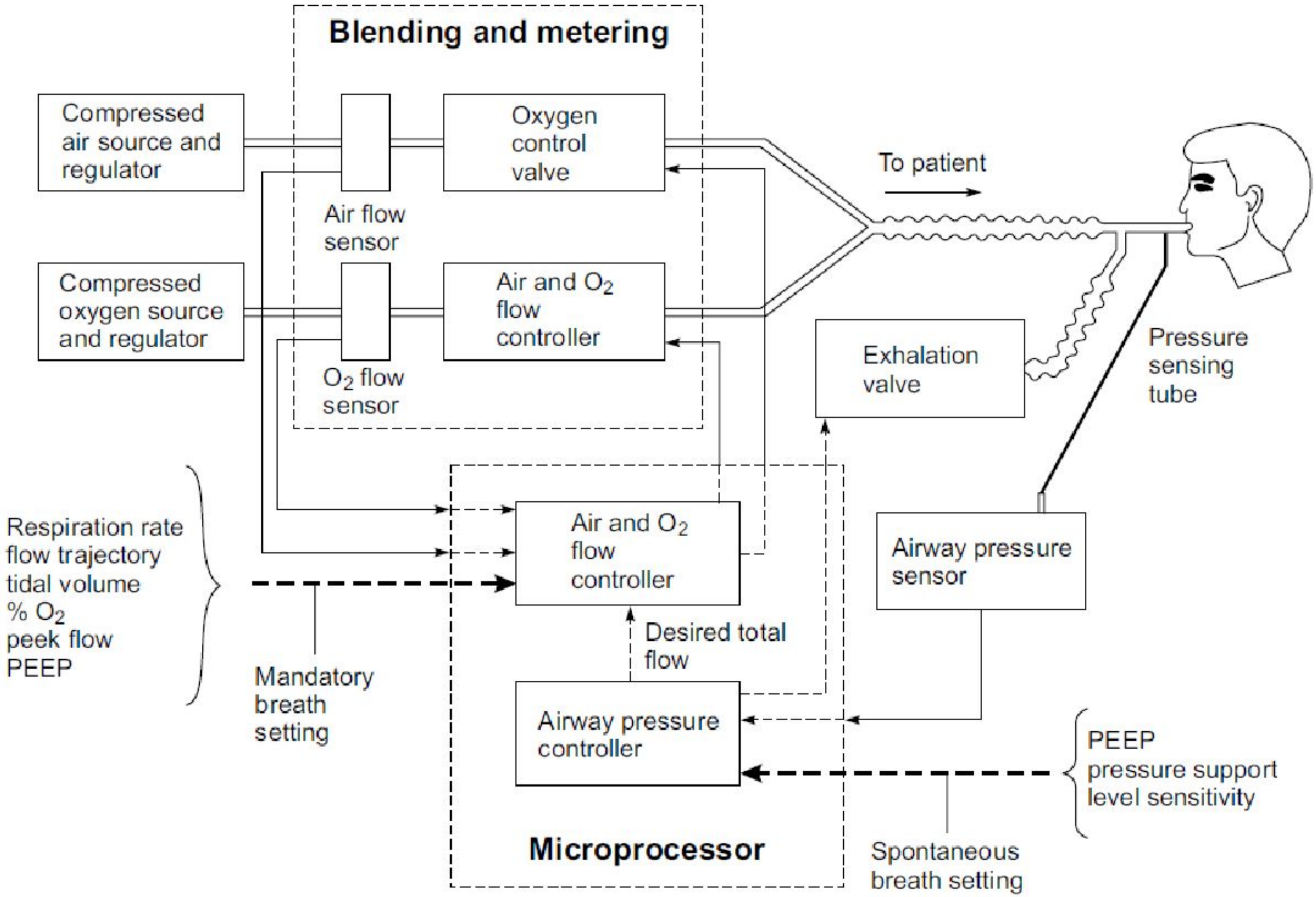
- Once inspiration has been triggered, inflation of the lungs continues until one of the following conditions occurs:
- The delivered gas reaches a predetermined pressure in the proximal or upper airways. A ventilator that operates primarily in this manner is said to be **pressure-cycled**.
- A predetermined volume of gas has been delivered to the patient. This is the primary mode of operation of **volume-cycled** ventilators.
- The air or oxygen has been applied for a predetermined period of time. This is the characteristic mode of operation for **time-cycled** ventilators.

BLOCK DIAGRAM OF VENTILATOR



CHAPTER

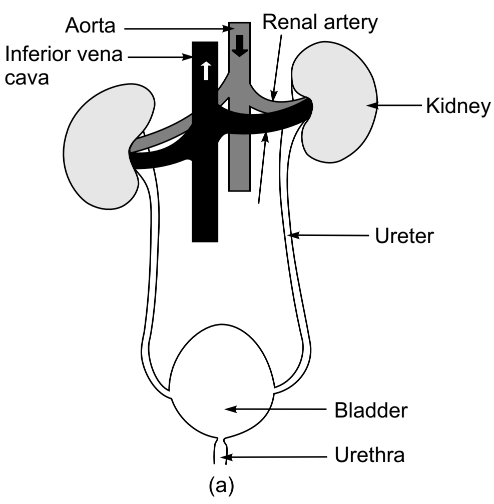
30

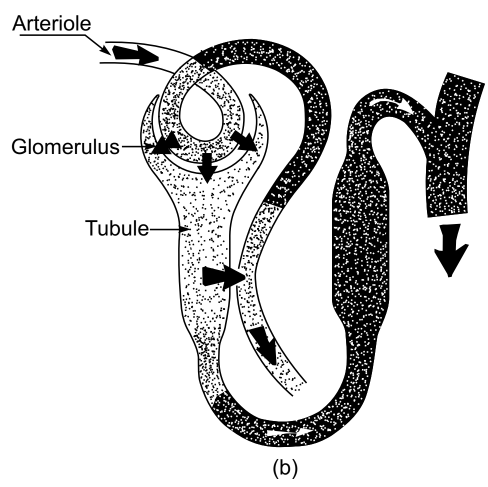
Haemodialysis Machines

30.1 FUNCTION OF THE KIDNEYS

The main function of the kidneys is to form urine out of blood plasma, which basically consists of two processes: (i) the removal of waste products from blood plasma, and (ii) the regulation of the composition of blood plasma. These activities not only lead to the excretion of non-volatile metabolic waste products but are also largely responsible for the remarkable constancy of the volume, osmotic pressure, pH and electrolyte composition of the extra-cellular body fluids.

The human body has two kidneys which lie in the back of the abdominal cavity just below the diaphragm, one on each side of the vertebral column [Fig. 30.1(a)]. Each kidney consists of about a million individual units, all similar in structure and function. These tiny units are called nephrons whose structure is shown in Fig. 30.1(b). A nephron is composed of two parts—a cluster of capillary loops called the glomerulus and a tubule. The tubule runs a tortuous course and ultimately drains via a collecting duct into the funnel-shaped expansion of the upper end of the ureter, i.e. the tube which conveys urine from the kidney to the bladder. The mechanism by which the kidneys perform their functions depends upon the relationship between the glomerulus and the tubule.





➤ Fig. 30.1 *(a) Kidneys and their connection to other organs of the body (b) structure of nephron*

The kidneys work only on plasma. The erythrocytes supply oxygen to the kidneys but serve no other function in urine formation. Each substance in plasma is handled in a characteristic manner by the nephron, involving particular combinations of filtration, re-absorption and secretion.

The renal arteries carry blood at a very high pressure from the aorta into the glomerular capillary tuft. The blood pressure within the glomerular capillaries is 70–90 mm of mercury. The blood flow through the capillary tuft is controlled by the state of contraction of the muscle of the arteriole leading to the tuft. The fluid pressure within the tuft forces some of the fluid part of the blood, by filtration, through the thin walls of the capillaries into the glomerulus and on into the tubule of the nephron. The glomerular filtrate consists of blood plasma without proteins. The total amount of glomerular filtrate is about 180 litres per day, whereas the amount of urine formed from it is only 1–1.5l. This means that very large amounts of water, and other substances, are re-absorbed by the kidney tubules. The re-absorption is partly an automatic process, because the absorption of water is accurately controlled by the anti-diuretic hormone of the pituitary gland, in relation to the body's need for water. The absorption of electrolytes such as sodium and potassium is partly controlled by the supra-renal gland and the concentration of others, like chloride and bicarbonate, is related to the acid-base balance. Some of the re-absorption from the glomerular filtrate is also a passive, automatic process of diffusion depending upon pressure gradients. This applies to water itself, and its electrolytes like sodium, potassium, chloride, calcium and bicarbonate.

There are other substances such as urea, phosphates and sulphates which are the waste products of metabolism. They are unwanted by the body. The tubules are selectively porous to substances of importance to the body and impermeable to the unwanted. Therefore, the unwanted substances cannot diffuse back into the plasma and thus a large proportion is excreted in the urine. As the filtrate passes down the tubules, the concentration of waste products rises steadily and the specific gravity of normal urine varies from 1.015 to 1.030 as compared with 1.010 for glomerular filtrate. Other substances in the glomerular filtrate such as glucose and amino acids show little tendency to diffuse through the tubular walls and are returned to the plasma by a process of active re-absorption.

The total blood flow through the kidneys is about 1200 ml/min. The total extra-cellular fluid amounts to about 15 litres. The blood plasma and the extra-cellular fluid are in equilibrium with each other and, therefore, an amount of blood equivalent to all the extra-cellular fluid can pass through the kidneys once every 15 minutes. The water and electrolyte content of the blood plasma and, therefore, indirectly of the extra-cellular fluid are closely controlled by the kidneys. Kidneys also play an important role in maintaining the acid-base balance.

30.1.1 Changes in the Body Fluids in Renal Disease

The symptoms and signs of profound renal malfunction are known as uremia, meaning urine in the blood. Since most urinary contents are water-soluble, they reach high concentrations in blood, and result in deranged body parts and their physiology.

Severe cases of acute renal failure are characterized by the virtual cessation of urine formation. It is obvious that the kidney cannot exert any appreciable excretory or regulatory function. In such patients, even when protein foods are omitted completely from the diet, the production of urea and other nitrogenous waste products continues because of the metabolic breakdown of the body's own tissue proteins. These products, when retained, result in a progressive rise in their level in the plasma. Furthermore, since acids are formed by the oxidation of the sulphur and phosphorus contained in protein, the plasma bicarbonate will fall. Partly because of the acidosis and partly because of cellular and tissue breakdown, potassium is liberated into the extra-cellular compartment and the plasma potassium level rises. Dangerous changes in the volume and distribution of body water may also occur in acute renal failure.

Chronic renal failure results in changes in the body fluids which occur due to a progressive decrease in the number of functioning nephrons. With the decrease in functional nephrons, the clearance of urea, creatinine and other metabolic waste products will decrease proportionally. In consequence, the plasma concentrations of these substances will rise. A reduction in the glomerular filtration rate (GFR) will also lead to an increase in the plasma concentrations of substances (e.g., phosphate and sulphate) which are removed from the blood by filtration and partially re-absorbed by tubules. However, since tubular absorption of these substances also declines with a falling GFR, both their clearances and their plasma concentrations usually remain unchanged until the GFR has fallen to about 25% of the normal rate. At this stage, there is also a fall in the plasma bicarbonate.

With the decline in the number of functioning nephrons, the kidney also becomes less effective as a regulatory organ. Accordingly, a patient with chronic renal failure is liable to develop either deficiencies or excess of body fluid water and sodium. However, the volume and sodium concentration of the extra-cellular fluids often remain virtually unchanged until the renal function is very seriously impaired.

Uremia is the clinical state resulting from renal failure. The signs and symptoms of uremia are extremely diverse and frequently appear to point to disease of other organs. Uremia affects every organ of the body as certain substances that accumulate in uremia are clearly toxic. Middle molecules with molecular weights of 1000-3000 are major contenders for the villainous role of the uremic toxin.

30.2 ARTIFICIAL KIDNEY

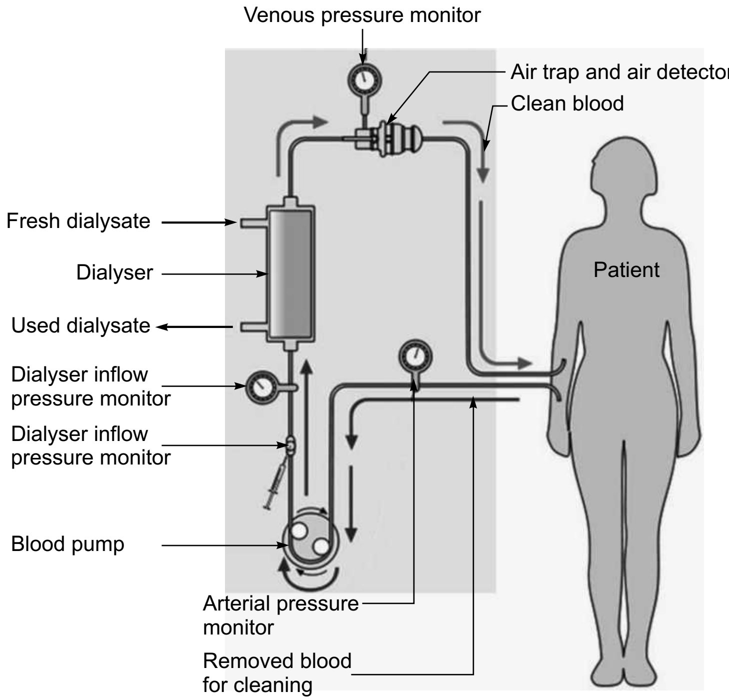
Artificial kidney attempts to replace the excretory functions of the kidney. Intermittent treatment with a mechanical device like the artificial kidney (dialyzer) will reduce the accumulation of waste products and water and thus the blood concentrations of the toxic substances are returned to normal levels. By effectively removing these materials from the blood, the dialyzer temporarily replaces the function of the natural kidneys and is able to keep the patient close to normal condition.

30.2.1 Types of Dialysis

Basically, there are two types of dialysis: haemodialysis and peritoneal dialysis.

30.2.1.1 Haemodialysis

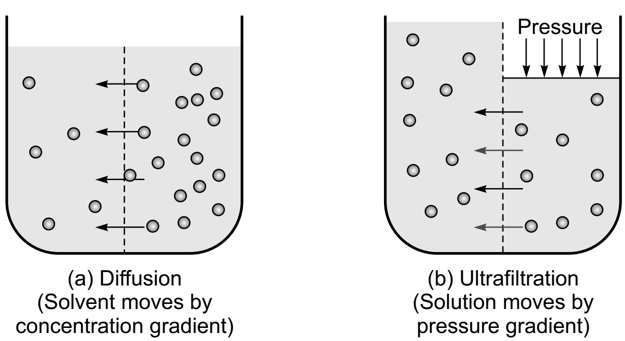
In haemodialysis, the patient's blood (Fig. 30.2) is pumped through the blood compartment of a dialyzer, exposing it to a semipermeable membrane. The cleansed blood is then returned via the circuit back to the body. Ultrafiltration occurs by increasing the hydrostatic pressure across the dialyzer membrane. This is usually done by applying a negative pressure to the dialysate compartment of the dialyzer. This pressure gradient causes water and dissolved solutes to move from blood to dialysate, and allows the removal of several litres of excess fluid during a typical 3 to 5 hour treatment.



➤ Fig. 30.2 *Principle of haemodialysis (http://www.renalmed.co.uk/database/haemodialysis-hd#)*

The main phenomena which take place in dialysis are:

Diffusion: It is the exchange of things dissolved in fluid (solutes) across the membrane due to differences in the amounts of the solutes on the two sides (concentration gradient). If there is a higher concentration of a given solute on one side of the membrane than on the other, then diffusion will occur to try to make the concentrations on both sides of the membrane the same. By controlling the chemicals in the dialysate, the dialysis machine controls this transfer of solutes across the membrane. Fig. 30.3 (a) shows the principle of diffusion.

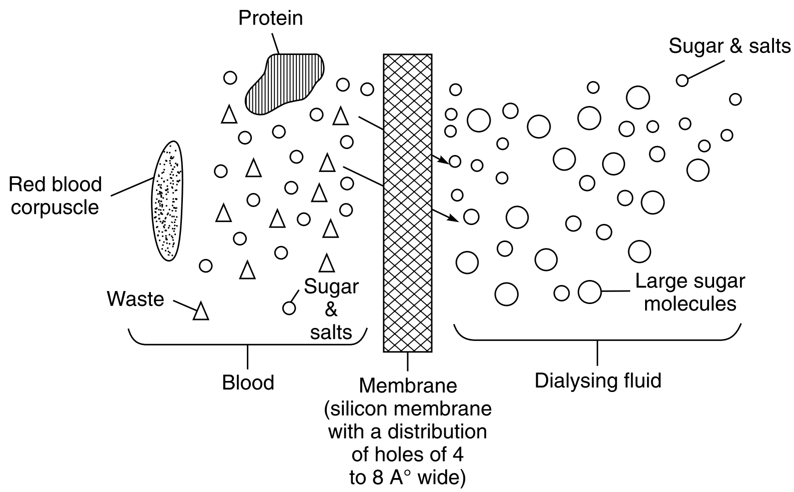


➤ Fig. 30.3 *(a) Diffusion: Solvent moves by concentration gradient (b) Ultrafiltration: Solution moves by pressure gradient*

Ultrafiltration: It is also referred to as convection, is fluid flow through the membrane, forced by a difference in pressure on the two sides of the dialyzer (pressure gradient). This controls the patient's weight loss over the course of the treatment. While earlier dialysis machines either controlled dialysate pressure or the pressure difference across the membrane in order to achieve ultrafiltration, modern dialysis machines are generally volumetric, meaning they control the volume of fluid removed from the patient directly and allowing dialysate pressure to change as it will in order to achieve the prescribed weight loss. Fig. 30.3 (b) shows the principle of ultrafiltration.

Essentially, an artificial kidney is a dialyzing unit which operates outside the patient's own body. It receives the patient's blood from the cannulated artery via a plastic tubing. The dialysate is an electrolyte solution of suitable composition and the dialysis takes place across a membrane of cellophane. The return of the dialyzed blood is by another plastic tube to an appropriate vein.

The dialyzing membrane has perforations which are extremely small (Fig. 30.4) and are invisible to the naked eye. Waste products in the blood are able to pass through these minute perforations into the dialysate fluid from where they are immediately washed away. In dialysis, the blood is exposed to dialysis solution across a selectively permeable membrane, which allows low molecular weight compounds to move freely between the two solutions, but restricts the passage of larger molecules. The perforations in the dialysis membrane have an average diameter of 50 Å with an estimated range of 30 Å to 90 Å. The waste products pass through the membrane because of the existence of a concentration gradient across the membrane. The dialysate fluid is free of waste product molecules and, therefore, those in the blood would tend to distribute themselves evenly throughout the blood and the dialysate.



➤ **Fig. 30.4** ***Function of dialyzing membrane in the artificial kidney***

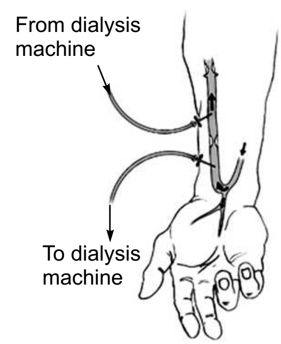
This movement of waste product molecules from the blood to the dialysate results in cleaning of the blood.

The volume of body fluid cannot be controlled by dialysis. Instead, ultra-filtration across the membrane is employed. For this, a positive pressure is applied to the blood compartment or a negative pressure established in the dialysate compartment. Either way, fluid—both water and electrolytes—will move from the blood compartment to the dialysate, which is subsequently discarded. The degree of ultra-filtration depends both on the pressure difference across the membrane and the ultra-filtration characteristics of the membrane.

The artificial kidney is thus simply a membrane separation device that serves as a mass exchanger during clinical use. It is unable to perform any of the synthetic or metabolic functions of the normal kidney and, therefore, cannot correct abnormalities that result from the loss of these functions. The only use of the artificial kidney in replacing renal function, therefore, is the transfer of noxious substances from the blood to the dialysate, so that they might be eliminated from the body. The physical processes by which solutes move across the dialysis membrane are discussed by Henderson (1976).

The application of repeated dialysis as a definitive treatment of the permanent loss of kidney function requires a simple method for obtaining repeated access to the patient's circulation over a period of years. This is achieved either by the Scribner Teflon-Silastic arterio-venous shunt or by using a subcutaneous arterio-venous fistula, Fig. 30.5 surgically created by anastomosis of the radial artery to an adjacent branch of the cephalic vein in the forearm. The fistula can then be used for dialysis by insertion of wide-bore needles through the skin into the veins. The fistulas have proved to be essentially non-clotting, rarely become infected and help to keep the patient free from an external device between dialyses.

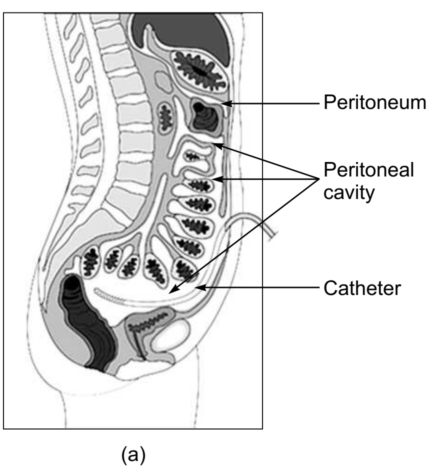
The maximum number of dialyses is performed for the treatment of chronic renal failure. This is maintenance dialysis, replacing the excretory functions of the kidney for the duration of the patient's life, or until he receives a transplant. Acute renal failure is the next major indication. Dialysis for acute renal failure is necessarily undertaken only for a few days or weeks until the patient's own kidneys recover.

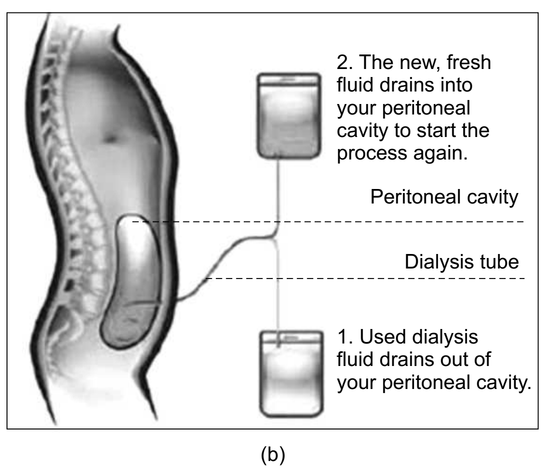


➤ **Fig. 30.5** *Arteriovenous Fistula*

30.2.1.2 Peritoneal Dialysis

Peritoneal dialysis (PD) works inside the body, using the peritoneal membrane, or abdominal lining, as a natural filter to remove waste from the bloodstream. In this form of dialysis, blood never leaves the body. Dialysis fluid enters the peritoneal cavity [Fig. 30.6(a)] through a small, plastic





➤ **Fig. 30.6** ***(a) Position of peritoneal cavity in the abdomen (b) peritoneal dialysis process (http://www.mykidney.org/TreatmentOptions/PeritonealDialysis/PeritonealDialysis.aspx)*** tube, called a catheter, surgically inserted in the abdomen. Extra fluid and waste travels across the peritoneal membrane into the dialysis fluid, which is then drained from the abdomen [Fig. 30.6(b)].

There are two types of PD therapy, automated peritoneal dialysis (APD), primarily performed at home while a patient sleeps; and continuous ambulatory peritoneal dialysis (CAPD), also primarily a home-based therapy, that provides continuous dialysis, 24 hours-a-day.

30.3 DIALYZERS

The dialyzer is the part in the artificial kidney system in which the treatment actually takes place and where the blood is freed from the waste products. It is the meeting point of two circuits, one in which the blood circulates and the other in which dialysis fluid flows.

Dialyzers, in routine clinical use, may be classified according to two basic design considerations: parallel plate and hollow fibre type. Each type of dialyzer has certain optimum operating requirements. The rate of clearance of substances such as urea, creatinine, etc. from the blood during passage through an artificial kidney is dependent upon the rate of the blood flow. As the flow rate falls, there is a disproportionate fall in clearance. At high flow rates, there is little advantage in further augmentation of the blood flow. The rate and pattern of the dialysate flow also influence overall performance in respect of clearance of waste products. Almost all commercial dialyzers use cellulosic type membranes, the most common being Cuprophan (cupro-ammonium regenerated cellulose).

The removal of waste products during dialysis is proportional to the concentration gradient across the membrane. In order to affect the maximum gradient, the concentration of waste products in the dialysate should be maintained at zero. This is achieved in most currently employed machines by using the dialysate only once and then discarding it. In addition, counter-current flow through the artificial kidney is used so that the dialysate enters the kidney at the blood exit-end where blood concentration of waste products is at the lowest level.

It is desirable for the resistance to blood flow in the dialyzer to be as low as possible, eliminating the need to employ a blood pump. In addition, the design of the blood compartment should be such that all the blood can be easily and completely returned to the patient at the end of dialysis. The design must affect an optimum, thin film of blood going through the dialyzer without streaming under perfused areas of membrane surface. Similarly, there must be optimum mixing in the dialysate compartment, effected via the membrane support structure.

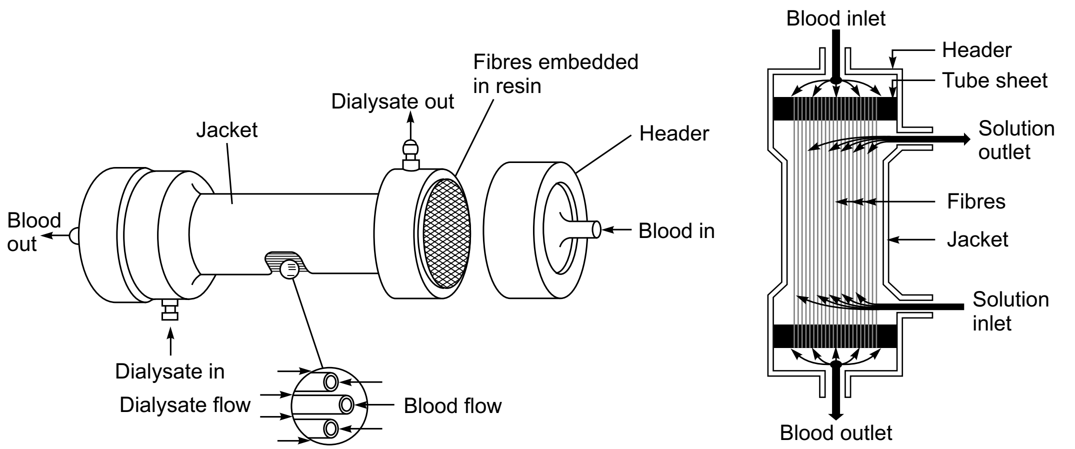
30.3.1 Parallel Flow Dialyzers

The KIIL dialyzer (Parallel plate dialyzer) has earlier been the most commonly used form of parallel flow dialyzer. It consists of three polypropylene boards with dialyzing membranes laid between them. The boards are held firmly with a frame on the top and bottom and are fastened by a series of bolts on the side. A rubber gasket runs along the periphery of the board's inner surface to prevent blood and dialysate leakage. The dialysate enters through a stainless steel port and is distributed to grooves running across the end of the board both above and below the membrane of each layer. After flowing down longitudinal grooves in the boards, it is collected and flowed out at the opposite end of the board. The outside measurements of a KIIL dialyzer are $125 \times 40 \times 16.5$ cm.

The KIIL dialyzer is not disposable. It needs to be cleaned and re-built after each dialysis operation. With this type of dialyzer, a single-pass body temperature dialysate passes through the dialyzer once before going to the drain to obtain higher operational efficiency and to minimize bacterial infection. This type of dialyzer is no longer in use.

30.3.2 Hollow Fibre Haemodialyzer

The hollow fibre haemodialyzer [Fig. 30.7 (a)] is the most commonly used haemodialyzer. It consists of about 10,000 hollow de-acetylated cellulose diacetate capillaries. The capillaries are jacketed in a plastic cylinder 18 cm in length and 7 cm in diameter. The capillaries are sealed on each end into a tube sheet with an elastomer. The capillaries range from 200–300 mm internal diameter and a wall thickness of 25–30 µm. The dialyzing area is approximately 9000 cm^2/unit. The primary volume with blood manifolds exclusive of tubing is approximately 130 ml. The blood is introduced and removed from the haemodialyzer through manifold headers. The dialysate is drawn through the jacket under a negative pressure around the outside of the capillaries counter-current to the blood flow. The dialyzers are disposable. Fig. 30.7 (b) shows internal details of hollow fibre dialyzer.



(a)
Constructional details of hollow-fibre dialyser

(b)
Hollow fibre dialyzer – internal details

➤ **Fig. 30.7** ***(a) Constructional details of hollow-fibre dialyser. (b) Hollow fibre dialyzer – internal details***

Disposable dialyzers offer the advantages of a reduction in infection risk and reduced operator set-up time. The dialyzer sterilization procedure also gets eliminated. However, the use of disposable dialyzers is an expensive procedure. This has necessitated the development of 'methods of cleaning dialyzer cartridges so that they may be re-used' (Deane *et al.*, 1978; Wing *et al.*, 1978). However, there are several difficulties associated with the practice of dialyzer re-use. It is a time-consuming and unpleasant procedure and requires additional technical skills.

30.3.3 Performance Analysis of Dialyzers

The dialyzer performance can be compared in terms of their clearance of urea and creating priming volume, residual blood volume, ultra-filtration rate, convenience of handling and cost, etc.

Clearance: The overall performance of the dialyzer is expressed as the clearance, analogous to that of a natural kidney. It represents that part of the total blood flow rate through the dialyzer which is completely cleared of solute. Uremic patients carry a number of toxic solutes in their blood, which are generated daily. Despite the uncertainty as to which solutes and how much should be removed, the performance of a dialyzer is generally assessed for a spectrum of molecular weight solutes. The molecular weight of urea is 60, of creatinine, 113, Vitamin B_{12}, 1355 and of insulin, 5200.

The clearance of urea and creatinine is measured at clinically useful blood flow rates and standard dialysis fluid addition or flow rate. It is calculated as

$$\text{Clearance} = \frac{\text{blood flow rate}}{\text{A} + \text{B} \times \text{blood flow rate}}$$

(where A and B are constants) with 95% confidence limits of the mean by least square approximation. The blood flow rate is measured by bubble transit time over a two-meter track using the mean of three measurements. Urea and creatinine concentrations are measured in the plasma from 1 ml sample of heparinised blood. Usually, the blood flow is maintained between 75–300 ml/min and dialysis fluid flow rate at 500 ml/min.

The performance of hemodialyzers is usually compared by employing the dialysance curve which is a graph of dialysance versus the blood flow rate.

$$\text{Dialysance D} = \frac{Q_b \ (C_{bi} - C_{bo})}{C_{bi} - C_{di}} \tag{1}$$

where Q_b = blood flow rate

C_{bi} = blood solute concentration at the dialyzer inlet

C_{bo} = blood solute concentration at the dialyzer outlet

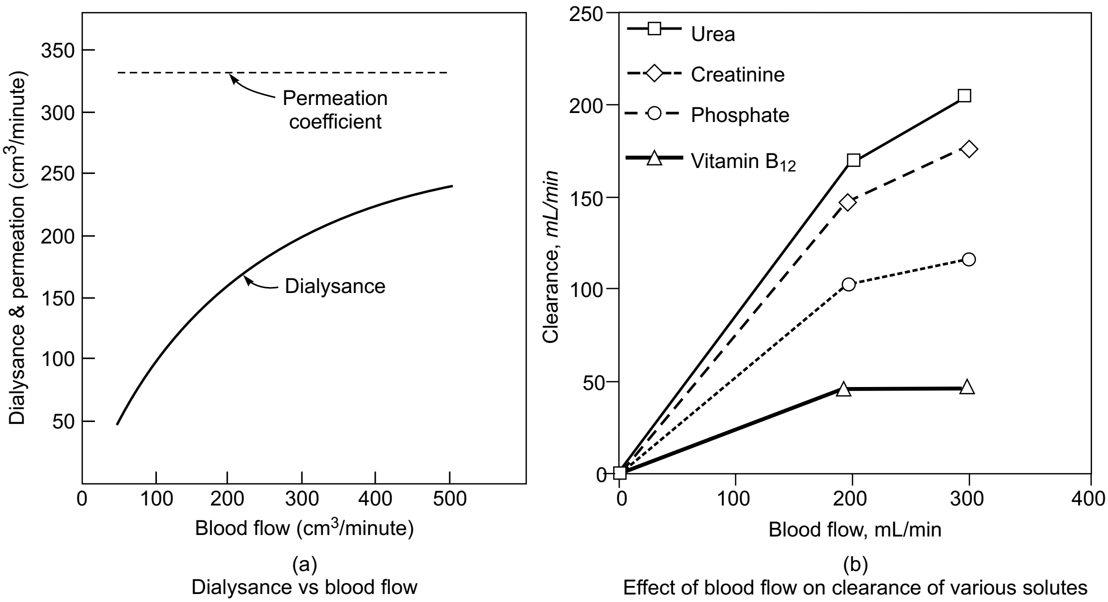
C_{di} = dialysate solute concentration at the inlet.

Fig. 30.8 (a) and equation (1) show that dialysance rises rapidly at low blood flow rates and tends to stabilize at high flow rates. Therefore, for comparing the performance of the dialyzer, it is important to specify the blood flow at which the measurements are made. The equation also shows that more waste is likely to be removed early in the treatment when the blood dialysate gradient is high.

The dialysance is also used to calculate the clearance, by the following relationship:

$$\text{Clearance} = \frac{\text{Dialysance}}{1 + \dfrac{\text{Dialysance}}{\text{dialysis fluid addition rate}}}$$

It may be noted that the clearance may vary with time despite quasi-steady state conditions. If the dialysate is re-circulated, its solute concentration increases, which effectively reduces the concentration driving force. For a given blood flow rate, the clearance is greater for the smaller molecular sized constituents. Fig. 30.8 (b) shows effect of blood flow on clearance of various solutes. From the figure, it is evident that small molecules such as urea (molecular weight 60D) are cleared more easily than large molecules such as vitamin B_{12} (molecular weight 1355 D). This is due to less membrane resistances and higher liquid diffusion coefficients for smaller-molecular weight solutes. On the other hand, the contribution of the membrane resistance to the overall value becomes greater with the increase in solute molecular weight. Bobb *et al.* (1971) suggested 'square metre hour' concept for increased removal of middle molecules by either increasing dialysis time or increasing membrane area. They illustrated that inadequate removal of solutes of 300–2000 daltons (middle molecules) might be associated with the neurological dysfunction in chronic uraemic patients. Keeping in view this study, attempts have been made to design dialyzers with large surface area (2.5 m^2) and development of more permeable membranes. Another method of comparing the performance capacity of dialyzers is given by:



➤ **Fig. 30.8** ***(a) Dialysance vs blood flow (b) Effect of blood flow on clearance of various solutes***

$$\text{Performance capacity} = K_s A = \frac{A}{R_s} \tag{2}$$

where A = surface area

K_s = permeability coefficient

R_s = mass transfer resistance, i.e., reciprocal of permeability.

Mass transfer resistance is composed of resistance due to a blood film layer, resistance of the membrane itself and the dialysate film resistance layer. Equation (2) is independent of blood flow rate and can be considered as a measure of dialyzer performance.

Ultra-filtration Rate: The fluid removal during dialysis (ultra-filtration) takes place due to hydro-static and osmotic transmembrane pressure gradients. The rate of fluid removal due to hydrostatic pressure effects depends upon the specifications of the dialyzer in terms of mass-transfer coefficient and surface area. It, however, has a linear function of the transmembrane pressure (TMP) gradient. It is given by

$$\text{Mean transmembrane pressure} = 1/2\,[P_{BI} + P_{BO}] - 1/2\,[P_{DI} + P_{DO}]$$

where P_{BI} and P_{BO} are the blood inlet and outlet pressures and P_{DI} and P_{DO} are the dialysate inlet and outlet pressures respectively.

The pressure losses generated by blood and dialysate flows in their respective flow paths should be small. This ensures that the local transmembrane pressure (ΔPm) will not vary excessively from the mean pressure. High values of ΔPm can result in deformation of the membrane and its possible rupture.

In actual clinical practice, only one blood pressure and one dialysate pressure are normally measured. Therefore, in order to obtain reasonable control over ultra-filtration, the pressure loss in the blood and dialysate compartments should be known as a function of the respective flow rates. The pressure drop (difference between blood inlet and outlet pressures, ΔPb) across a dialyzer is directly proportional to the length of the passage and the viscosity of the fluid and inversely proportional to the number of blood passage and some function of their cross-sectional area. Blood passage width may change with changing blood compartment pressures. Therefore, the relationship of pressure drop to blood flow is not linear at increased flows which are accompanied by increased pressures that can cause a widening of the blood passage and a decrease in DP_b/Q_b. The viscosity of the blood is not constant as the blood is an anomalous fluid. Its viscosity tends to increase with haematocrit and decrease in small passage. The area of cross-section is important as with small areas, an inordinately high pressure is required to yield a given flow or flow is reduced to very low levels for a given pressure source.

Residual Blood Volume: Residual blood volume can be measured after an 800 ml saline wash in. The fluid remaining in the dialyzer and lines is circulated through a 0.1 litre bottle of 0.04% ammonia solution for 10 mm. The residual blood volume is calculated from the formula.

$$\text{Residual blood volume} = \frac{\text{U (1000 + volume of dialyzer and lines in ml)}}{\text{200 S}}$$

where U = the haemoglobin concentration of the re-circulated fluid,

S = the haemoglobin concentration of a sample of arterial blood taken at the end of dialysis and diluted 1: 200 with 0.04% ammonia.

Residual volumes of 1.8 to 6.3 ml are quoted in the literature depending upon the dialyzer type and washback volume (Hartitzsch *et al.*, 1973).

Priming Volume: The volume of blood within the dialyzer is known as priming volume. It is desirable that this should be minimal. Priming volume of present day dialyzers ranges from 75 to 200 ml, depending on the membrane area geometry and operating conditions. Requirement of low priming volume permits the use of patients own blood to prime the circuit without serious hypovolemic effects. This is particularly significant in the case of long-term dialysis therapy.

Extra-corporeal blood volumes become important in those dialyzers which require priming. Priming is usually accompanied at relatively low pressures. Recent innovations have considerably reduced the extra-corporeal volume and a saline prime is frequently used.

Pyrogenicity: Pyrogen reactions are rare with all disposable dialyzers. However, they are known to exist with KIlL dialyzers but at rates well lower than 1%.

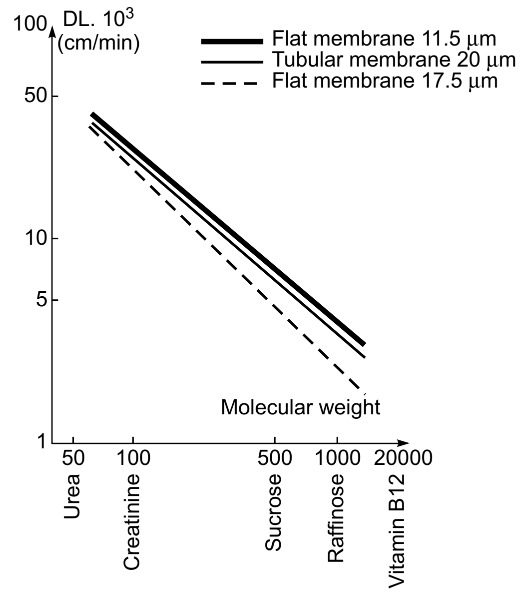
Leakage Rate: Blood-to-dialysis-fluid leak with the KIlL dialyzer is found to be 3% (Rastogi *et al.*, 1969), but it varies with the dialyzer, the batch of membrane and the skill of the operator. The leak rate from all cuprophan coils is, however, high (Easterling *et al.*, 1969).

30.4 MEMBRANES FOR HAEMODIALYSIS

The efficiency of dialysis is determined by the permeability characteristics of the semi-permeable membrane. The ideal membrane should possess high permeability to water, organic metabolites and ions, and the capability of retaining plasma proteins. The membrane should be of sufficient wet strength to resist tearing or bursting and non-toxic to blood and all body cells. Since a fresh membrane is required for each dialysis process, it should be inexpensive to produce.

Virtually all artificial kidneys presently in use, employ cellulosic membranes. Such membranes operate as sieve-type membranes allowing the passage of solutes through micro-holes. Therefore, selective sieving of the blood is based upon the size, shape and density of the solute.

Cupraphan (trademark of Enka Glanzstoff, Germany) is the commonly used membrane for haemodialysis. It is a membrane consisting of natural cellulose and is considered puncture-proof, and of high tenacity and elasticity. During haemodialysis, different substances of varying molecular weight are to be removed. The specific membrane permeability values and their dependency on substances with increasing molecular weight are shown in Fig. 30.9. The good permeability on the substances with middle molecular weight is obvious and substances with molecular weights of over 10,000 are retained.



➤ **Fig. 30.9** ***Relationship between permeability and molecular weight for different thicknesses of the membrane***

For applications in hollow fibre dialyzers, cupraphan fibres are used. With their small internal diameter, it is possible to design dialyzers with a low priming volume, making it possible to combine a large surface area with a low priming volume. The number of fibres in a dialyzer can be as high as 16,000 giving a density of fibres as 1000 per cm^2. Cupraphan hollow fibres are particularly suitable for the dialyses of solutes in the middle molecular weight range (500–2000). The high solute permeability in this range is not associated with an extremely high water permeability.

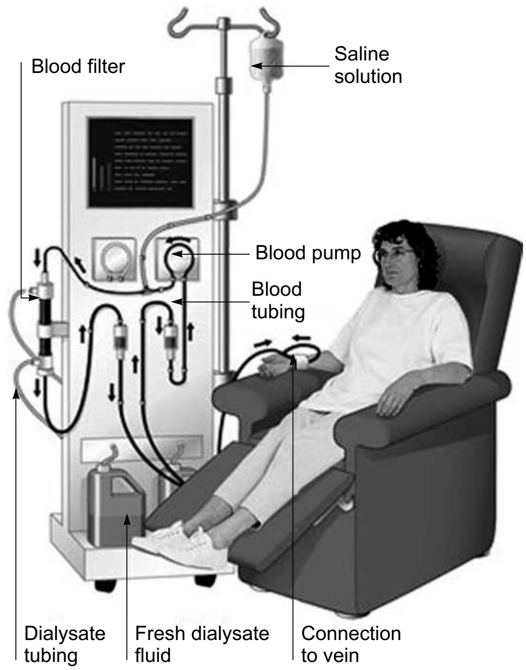
30.5 HAEMODIALYSIS MACHINE

A haemodialysis machine is used for the production of warm dialysate which is then circulated through an external dialyzer assembly. It also controls the cycling of the blood from the patient through the artificial kidney (dialyzer) and back to the patient. It continuously monitors and controls all important parameters, automatically halting treatment in the event of parameters going out of pre-set limits. The haemodialysis machine performs five basic functions. It (i) mixes the dialysate, (ii) monitors the dialysate, (iii) pumps the blood and controls administration of anti-coagulants, (iv) monitors the blood for the presence of air and drip chamber pressure, and (v) monitors the ultra-filtration rate. Fig. 30.10 shows a typical haemodialysis machine set up.

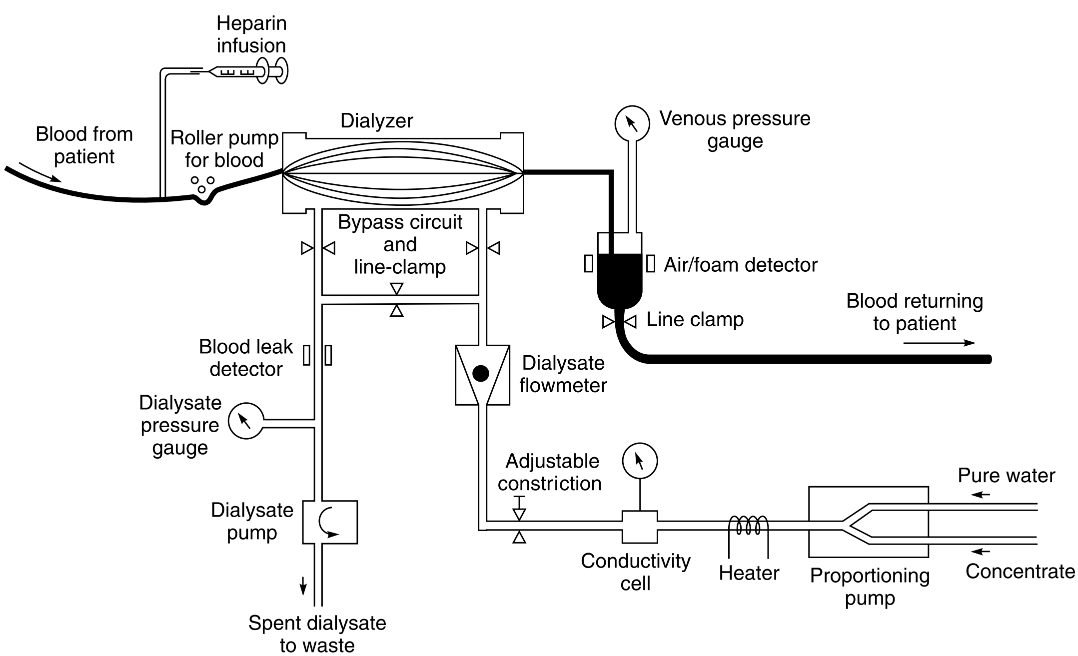
The machines are designed to be totally adjustable to meet individual therapy requirements. The machine pumps and controls the flow of blood from the patient through the dialyzer at a pre- determined rate and pressure to ensure effective clearances and fluid removal in a specified time period. Some machines also provide an ultra-filtration rate meter that measures the ultra-filtration rate in kilograms per hour. This allows the operator to efficiently and accurately calculate, predict and control fluid removal during dialyses. Fig. 30.11 shows a block diagram of a haemodialysis machine.

Proportioning Pumps: Mixing large amounts of dialysate from dry chemicals is time consuming and laborious. When re-circulated, glucose-containing dialysate results in the rapid growth of bacteria unless changed at frequent intervals. Single-pass proportioning systems using liquid concentrate avoid bacterial overgrowth. Proportioning systems of earlier designs used motor-driven positive displacement pumps. Water and concentrate pumps were driven by one shaft from the same motor, simultaneously delivering a fixed ratio (35:1) of water and concentrate into a mixing chamber.

Incoming water under controlled pressure can drive proportioning pumps, thus eliminating the need for a motor and permitting a smaller, quieter system. Incoming water activates a piston with an attached shaft, causing expulsion of a fixed proportion (35:1) of water (piston) and concentrate (shaft) from different chambers. A steady flow of dialysate is achieved by



➤ Fig. 30.10 *A typical haemodialysis machine set up*

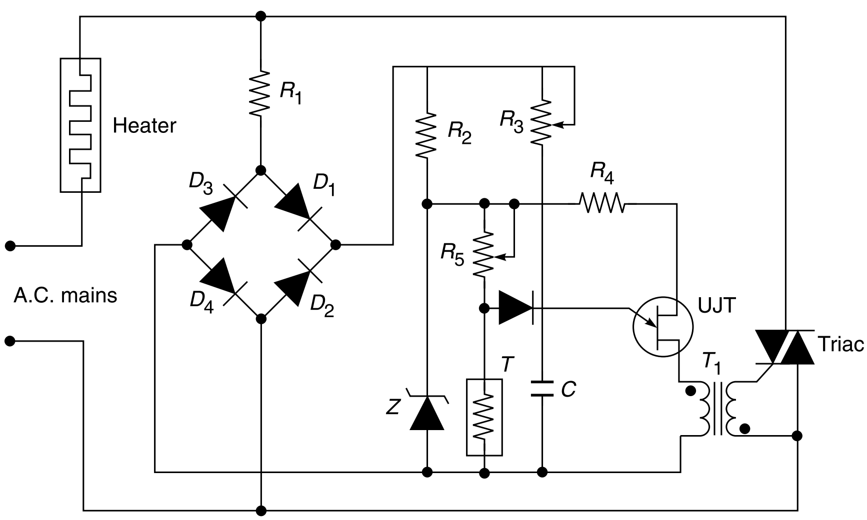


➤ Fig. 30.11 *Schematic diagram of haemodialysis machine. Blood flow path is shown in solid black line and the dialysate circuit as open lines* adding an additional concentrate chamber to the other side of the water chamber so that the back and forth movement of the piston and shaft expels water and dialysate continuously.

The proportioning system could be of the fixed ratio or variable ratio type. In a fixed ratio type, the ratio of concentrate and water is determined by the ratio of two pump piston diameters and these are mechanically fixed in their relationship. Proportioning is, therefore, fixed also, and is generally 34:1. In the variable ratio type system, a variation of ± 5% on the standard 34:1 dilution ratio is possible.

Dialysate Temperature Control and Measurement: The dialysis is normally done at the body temperature. The temperature of the dialysate is, therefore, monitored and controlled before it is supplied to the dialyzer. In case the dialysate gets over-heated, the system should stop the flow to the dialyzer and pass it to the bypass. Dialysis at temperatures lower than the body temperature is less efficient and requires re-warming of the blood before its return to the patient's body. Temperatures in excess of 40°C tend to damage components of the blood. A temperature control system is used to raise the temperature of the dialysate to the required temperature which can be varied from 36 to 42°C. A secondary safety cut-out ensures that the heaters are switched off if the temperature exceeds 43 °C.

For effecting control of temperature, proportional controller which makes use of a thermistor for sensing the temperature and a triac for control of power to the heater is used. A typical circuit for such a system is shown in Fig. 30.12. Normally, the uni-junction transistor is 'off' until the capacitor C charges to a point of breakdown voltage. When this occurs, the transistor conducts and the capacitor is discharged through the pulse transformer T. The triac thus gets a triggering pulse and switches on the heaters. The triac switches off at the end of each half-cycle and remains so until triggered once again. Since a triac conducts in both directions, it can be switched on during each half-cycle.



➤ **Fig. 30.12** ***Simplified circuit diagram for controlling dialysate temperature***

The thermistor has a negative temperature coefficient. With an increase in temperature from the set value, its resistance decreases, thereby reducing the rate of charge of C. Therefore, the frequency of charge and discharge (oscillations) reduces and less power is delivered to the heaters which results in a reduction in temperature. With this method, it is possible to control the temperature with an accuracy of 0.2°C. The temperature can also be controlled by varying resistance R_3 and therefore, any temperature can be set with the help of this control.

A thermistor connected in one arm of a Wheatstone bridge may be used as a sensor of temperature of the dialysate in the header tank. The output of the bridge can be amplified in a differential amplifier and displayed on a panel meter. The amplified signal would also operate alarm circuits in case the temperature of dialysate crosses the preset limits.

Some machines have facilities for automatic sterilization. Sterilization is carried out by passing water at 85°to 90°C, through the total hydraulic system. In these machines, sequence interlocks ensure that a dialysis cannot be started without sterilization and that minimum requirements, for adequate sterilization are achieved before the dialysis phase can be selected.

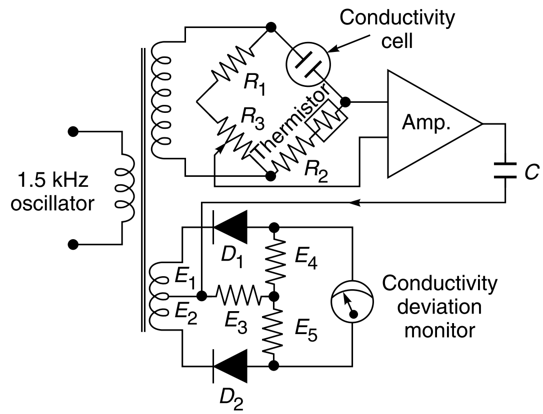
In the modern microprocessor-based haemodialysis machines, the temperature monitor and control circuitry generate a signal that the CPU utilizes to generate display of the fluid temperature and to control the heaters. A dual element heater assembly, which has a 150 W and a 300 W element, is used to bring the fluid up to and to maintain the operating temperature. When the fluid temperature rises to within 2.5°C of the preset temperature (between 35°C to 39°C), the 300 W heater is turned off and the 150 W heater is used to maintain the set temperature.

As a secondary fault monitor, the system incorporates a mercury type temperature switch which is normally open when the dialysate temperature is below 40°C (± 0.5°C) and closed when the dialysate temperature exceeds 40°C ((± 0.5°C). In case the fluid temperature exceeds 40°C, the microprocessor removes the heater elements. In addition to the thermistor probe and temperature switch, the enabling of the heaters is dependent upon the flow rate. The microprocessor reads the flow pulses and determines if there is adequate flow within the system. If the flow is inadequate, the heater elements are disconnected.

The ***flow*** is measured using a flow-thru transducer which produces a precise number of pulses per unit of flow (26,000/litre or 108 pulses/second at 250 ml/min). This is achieved by monitoring the rotation of a disk which contains light reflective white spots. Light pulses from the rotating disk are transmitted by internal fibre optics. The sensor assembly includes a light source and a photo-transistor to provide the optical coupling with the sensor. The pulses generated by the flow transducer are amplified, filtered and counted to determine the flow rate. These pulses are also used to control the flow rate in the hydraulic circuit. This circuit supplies a computer-generated variable drive signal to the dialysate pump and flow rate feedback signal to the CPU.

Conductivity Measurement: The conductivity of the dialysate being produced is continuously monitored by a conducting cell, to verify the accuracy of proportioning. The result is usually displayed as a percentage deviation from the standard. In practice, a fluctuation about the mean reading will occur and conductivity will normally be maintained with 1 %. If the conductivity not remaining within limits, an alarm is given. The effluent pump motor will be switched off automatically which effectively prevents further circulation of dialysate through the dialyzer, and dialysate production will be by-passed to the drain.

The composition of the dialysate is checked by comparing the electrical conductivity of the dialysate with a standard sample of the dialysate. Proper temperature compensation is essential as the conductivity of the dialysate changes by about 2% for every 1°C change in temperature. Fig. 30.13 shows a simplified block diagram of the conductivity measuring system. It comprises a 1.5 kHz oscillator which drives a bridge circuit, one arm of which contains a conductivity cell. The conductivity cell is of the flow type and is mounted directly downstream of the header tank. In order to provide a fast response to changes of solution temperature which would otherwise considerably affect the conductivity measurements, a temperature compensation thermistor is placed in another arm of the bridge. The output from the bridge, after amplification, is capacitively coupled to a phase-sensitive detector where its phase is compared with the phase of the 1.5 kHz oscillator output. The magnitude and phase of the output from the phase-sensitive detector determine the direction and amount of deviation from the pre-set value. Table 30.1 gives typical dialysate composition.



➤ **Fig. 30.13** ***Simplified circuit diagram for monitoring conductivity of dialysate***

• **Table 30.1** ***Composition of Dialysate***

Constituent	*Concentration (mEq/l) (g/l)*
Sodium (Na^+)	130.00
Potassium (K^+)	1.34
Calcium (Ca^{++})	3.30
Magnesium (Mg^{++})	1.00
Chloride (Cl^-)	101.00
Acetate	35.00
Lactate	1.30
Glucose	1.80

In practice, the mEq/l of sodium, calcium, chloride, potassium, magnesium and acetate are added to obtain the total ionic content of the dialysate in mEq/l. For example, for total ionic content of 270 mEq/l, the conductivity is 12.9 milli-ohms and for 304 mEq/l, it is 13.8 milli-ohms.

Dialysis must never commence unless it is known that the conductivity circuit calibration and concentrate in use are both correct for the intended dialysis. Therefore, it is recommended that once per month a sample of the dialysate from the machine's dialysate outlet connector be analyzed in a laboratory to check conductivity monitor calibration.

Dialysate Pressure Control and Measurement: Negative pressure upon the dialysate is created by the effluent pump. The effluent pump is a fixed flow, motor-driven gear pump. A small plastic housing encloses stainless steel gears driven by an electric motor. Pressures between zero and maximum are available by adjustment of a needle valve mounted on the machine panel. A relief valve (pre-set to suit the type of dialyzer) limits the maximum negative pressure available, thus minimizing the risk of a burst in the dialyzer membrane which may be caused by high transient pressures. Pressure adjustments should not produce any significant change in the flow rate. The pressure is measured by a strain gauge transducer connected immediately downstream of the dialysate return side. Pressures within the range 0 to –400 mmHg are generally made available and any value can be adjusted in this range.

The pressure indicated on the gauge will be the dialysate pressure on one side of the dialyzer membrane. On the other side of the membrane will be the venous pressure. The effective pressure across the membrane, which is so important in consideration of filtration and weight control, will be the algebraic sum of the dialysate pressure and venous pressure. If the pressure goes beyond the alarm limits, the effluent pump is switched off automatically and dialysate production by-passed to drain by way of the header tank overflow and the waste funnel.

Venous Pressure Measurement: Venous pressure is normally measured at the bubble trap. A length of tubing connects the trap to a small plastic housing to which a strain gauge transducer is attached. The elevation above the floor of the point of connection to the blood line produces a small change in the pressure reading. For maximum accuracy, the sensor connection should be maintained in the same altitude during dialysis, preferably with the luer connector downwards to prevent blood reaching the diaphragm in the event of a leak. If the venous pressure passes beyond one of the alarm limits, power to the blood pump will be isolated and the blood pump, if in use, will cease to operate.

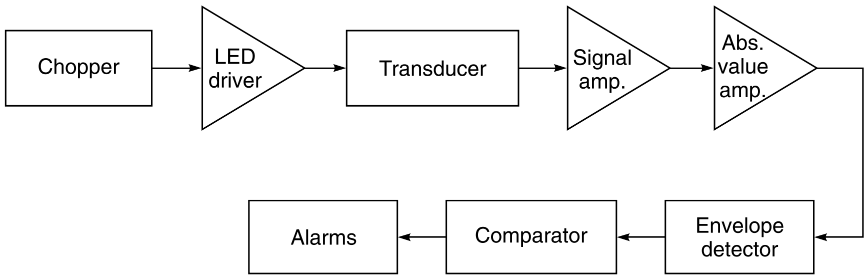
Blood Leak Detector: In a dialysis machine, a thin membrane separates the patient's blood from the dialysate. Normally, the pressure on the blood side of the membrane is maintained at a much higher level than the pressure on the dialysate side. This is necessary to minimize the time required for the dialysis procedure. Besides this, in order to reduce the total time required for dialysis, the membrane area is made as large as possible. Therefore, these two conditions of a high pressure differential across a large fragile membrane may result in a leak in the membrane. In fact, even a relatively small tear in the membrane can result in major blood loss in a very short time, with the consequent immediate threat to the patient.

Blood-to-dialysate leaks usually occur at the beginning of dialysis and can be detected by examination of the effluent from the dialyzer. However, it is not rare to have instances wherein blood leaks have been found to start several hours after setting up dialysis. The detection of blood leaking through imperfection in the membrane into the dialysate is best achieved by monitoring the effluent from the dialyzer for changes in transmission of light resulting from the presence of haemoglobin. If there is any blood leak across the dialyzer membrane, it can be detected by using a photo-electric transducer.

The dialysis membrane leak detector (Rhodine and Steadman, 1976) basically examines the light absorption of the dialysate at 560 nm, the absorption wavelength for haemoglobin. This spectral tuning of the system makes it sensitive and stable, and reduces false alarm situations. An LED is available which has a peak spectral emission at 560 nm with a spectral line half width of 27 nm.

Fig. 30.14 shows a block diagram of the blood leak detector. In order to minimize drift over a period of several hours required for the dialysis, a chopped light system with AC amplifiers is employed. Chopping is achieved by driving the LED with a square wave of current. The light is detected with a photo-detector. After amplification of the AC response signal, an absolute value circuit provides a signal whose peak value is proportional to the received 560 nm light. The peak value is compared to a reference voltage which is pre-set. When the peak value falls below the selected threshold, visual and audible alarms are activated.

Blood leak detectors are liable to give false alarms when used over a long period of several weeks. This is the result of a gradual build-up of contaminants on the lenses of the LED and the detector. This needs gradual change in the setting of the threshold. Careful cleaning of the



➤ Fig. 30.14 ***Block diagram of blood leak detector using LED as a light source (After Rohdine and Steadman, 1976)***

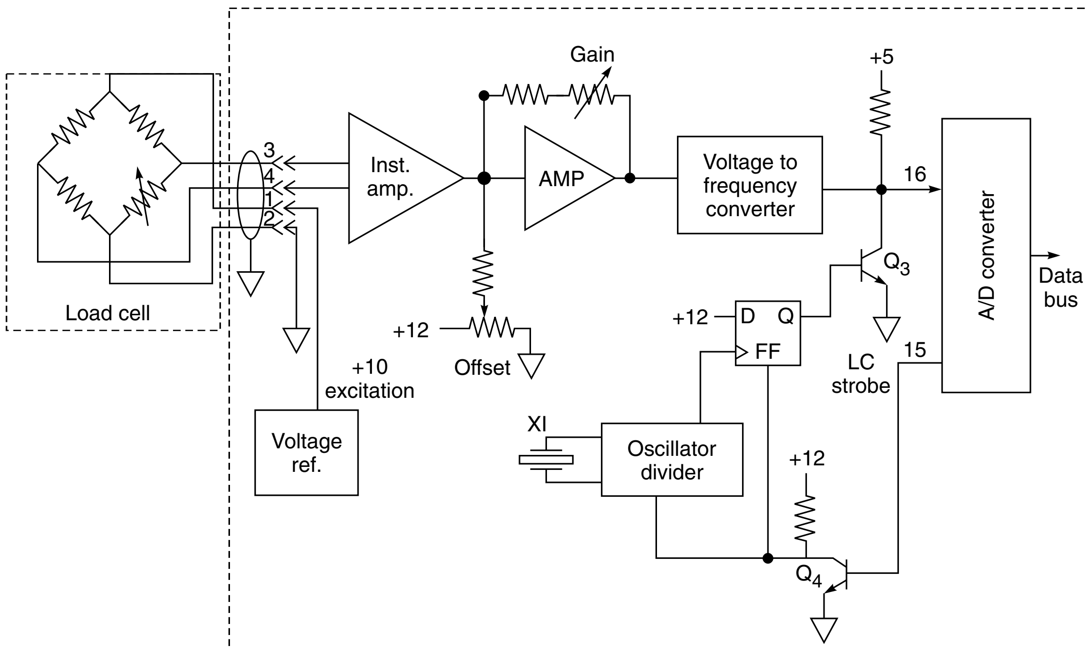
transducer can, however, restore the original threshold. This does not materially affect system performance since in almost all machines, the threshold is set at the beginning of each dialysis procedure.

Blood leak level, for normal operation, is set at 25 mg of haemoglobin per litre of dialysate. The maximum setting detects blood leaks at the rate of 65 mg/l of dialysate. If a blood leak is detected, the effluent pump is switched off automatically and dialysate production by-passed to drain by way of header tank overflow. The blood pump is de-energized and, if in use, ceases to operate.

Ultra-filtrate Monitor: The ultra-filtrate monitor circuit is used to monitor the amount of fluid removed from the patient and in conjunction with the negative pressure, to control the rate at which it is removed. This circuit generates a signal that the CPU utilizes to generate display of the total UF (ultra-filtrate). The CPU also uses this signal to calculate and maintain the UF rate required by the operator. If the calculated negative pressure is less than 30 mmHg or greater than 350 mmHg, the CPU will generate an alarm signal.

Fig. 30.15 is a block diagram of the ultra-filtrate monitor. The load cell and associated electronics are used to monitor the weight changes of the fluid in the reservoir during the haemodialysis treatment. The load cell utilizes a strain gauge that produces a differential resistance proportional to the applied force. An excitation of 10V DC is supplied to the strain gauge bridge from a reference source. The differential output from the strain gauge bridge is typically 13.3 mV for a 10 kg load. The differential input is first connected to the instrumentation amplifier which gives a gain of 100 and produces a single-ended output. The following amplifier stage includes provisions for offset and gain adjustments. The weight is represented at this stage by a DC voltage. It is changed to a proportional frequency by a voltage-to-frequency converter. The pulses corresponding to the weight are then counted (A/D Converter) and given to the microprocessor.

Other Sensors: Dialysis machines require many different types of sensors to monitor various parameters. Blood pressure at various points in the extracorporeal circuit, dialysate pressure, temperature, O_2 saturation, motor speed, dialyzer membrane pressure gradient, and air are all monitored for proper values during dialysis. Unless they have digital outputs, the sensors' analog signals must be amplified, filtered, and digitized before being sent to the microcontroller. The sensors require a range of ADC resolutions depending on the criticality of their accuracy, and a range of sampling speeds depending on the response times required. In the modern machines, microcontrollers are used for processing the digital signals before they are sent to the display.



➤ Fig. 30.15 *Block diagram of ultrafiltration monitor*

Flow Meter: The flow is measured with a flow meter which comprises a stainless steel float, inside a glass tube held by plastic connectors. All fluid returning to the machine from the dialyzer passes through the flow meter. The dialysate flow rate is fixed at a nominal 500 ml/min. Since it is generally positioned downstream of the dialyzer, it is possible to observe a large blood-to-dialysate leak by the discoloration of the fluid in the flow meter tube.

Effluent Pump: Effluent pumps are available in several design configurations. The more common types are: the diaphragm type, gear type and magnetically coupled. The diaphragm type pumps are not preferred because they give problems due to diaphragm fatigue when operated over long periods. Therefore, in the modern machines, either the gear type or magnetically coupled pumps are preferred.

Blood Pump: The blood pump used in dialysis machines is usually of the peristaltic type. It is designed to give blood flow at a rate of 50 to 350 ml/min. This type of pump is very convenient because it does not touch the fluids directly. Instead, a section of flexible tubing runs through the pump mechanism where it is compressed by rollers to push the fluid forward. These pumps require a significant amount of power and are driven by either DC or AC motors with variable speed control. Electronic means are provided to ensure that the motor is turning at the desired rate with the help of Hall-effect sensors. Peristaltic pumps are commonly used for driving the various higher volume fluids in the machine: blood, dialysate, water, and saline.

Valves: Several valves with electronic actuation are needed in the machine to allow variable mixing ratios. Various implementations are possible from simple opened/closed valves driven by solenoids to precision variable-position valves driven by stepper motors or other means.

Bubble Trap: Air embolism is a serious hazard in dialysis. Air may be sucked in due to inadequate flow in the line in the pumped dialysis system. Alternatively, air may be transferred from the dialysate. It is for this reason that dialysis equipment includes provision for adequate de-aeration of the dialysate. The venous return flow circuit usually incorporates a bubble trap to diminish the chances of air embolism. The level of blood in the venous return bubble trap may be monitored by a photo-electric cell. Some manufacturers use the ultrasound method for detecting the presence of air in the blood line.

Heparin Pump: These pumps are usually of the plastic syringe type, having a capacity of 30 cc. The delivery of heparin from the pump is calibrated in cm/h. The pump is driven by a stepper motor and a drive screw mechanism. This drives the plunger of the syringe into its barrel which produces the pumping action. The stepper motor speed is determined by the computer based on the heparin flow rate required. The machines are generally provided with the facility to accommodate commonly used syringe sizes. The speed of the stepper motor is monitored using an optical encoder.

Blood Pressure Monitor: The blood pressure monitor circuit is used to monitor the arterial and venous blood pressures. Two separate pressure transducers of the strain gauge type are used for this purpose.

***Microcontrollers*:** Because of the large number of input signals to be monitored and the large number of pumps and other mechanisms to be controlled, many of these functions are performed with dedicated microcontrollers for that portion of the system. Controlling the overall system will be a main processor capable of running a full operating system and graphical user interface (GUI). Communication between the controllers is required to send data, commands, and alerts.

Data Interfaces: A running record of the dialysis process for each patient session is kept electronically and made available in a number of ways. Dialysis machines can include USB, Ethernet, and a variety of serial (RS-232, RS-485, RS-422, etc.) interfaces to legacy hospital information systems. Wireless interfaces (such as Wi-Fi®) may also be included for direct connection to hospital wireless networks.

Data card slots are also available on some designs. This allows patients to carry an ID card with personal medical information stored on it to enable automatic setup of many of the machine parameters.

Cleaning System: Between patient sessions, any reused component is sterilized. One approach is to heat water or saline to a high sterilizing temperature and then run it through the machine, both through the extracorporeal circuit and through the dialysate circuit.

Fail-safe Circuitry: The equipment must contain comprehensive self-test and fault-indication capabilities, which require additional circuitry and the use of components that include self-test features. ICs with self-test and fault-reporting capabilities are very useful for maintaining patient safety under single-fault conditions. Additional monitoring circuitry is commonly used to monitor power-supply voltages, while watchdog circuits are used to ensure that microcontroller operation remains within bounds. Both audible and visible alarms are provided to alert users when a warning is needed or a fault condition has occurred.

Electrical leakage to the patient is a significant concern. The haemodialysis machine must meet the requirements of the IEC 60601-1 product safety standard for electrical medical equipment.

From an operational perspective, dialysis equipment must meet specific safety criteria. One of these criteria is single-fault tolerance, which means no single point-of-failure in the pumps, motors, tubes or electronics will endanger the patient. To achieve single-point tolerance, there must be several redundant components and circuits, as well as "watchdog" managed-disengage system mechanisms.

A safe mode of operation involves disabling the arterial blood pump and clamping the venous line to prevent unsafe blood from flowing to the patient. Both active and passive components, such as control devices, sensors, motors, heaters, pumps and valve drivers, are needed for this type of functionality.

Monitoring and control equipment forms an essential part of the haemodialysis system as it helps in maintaining the most optimum conditions during dialysis. In other words, it ensures a safe clinical procedure against any potential hazards. Incorrect composition and temperature of the dialysate, blood loss due to disconnection or membrane leakage and the formation of air embolism in the blood are some of the principal hazards which, when developed, require immediate remedial steps for the safety of the patient.

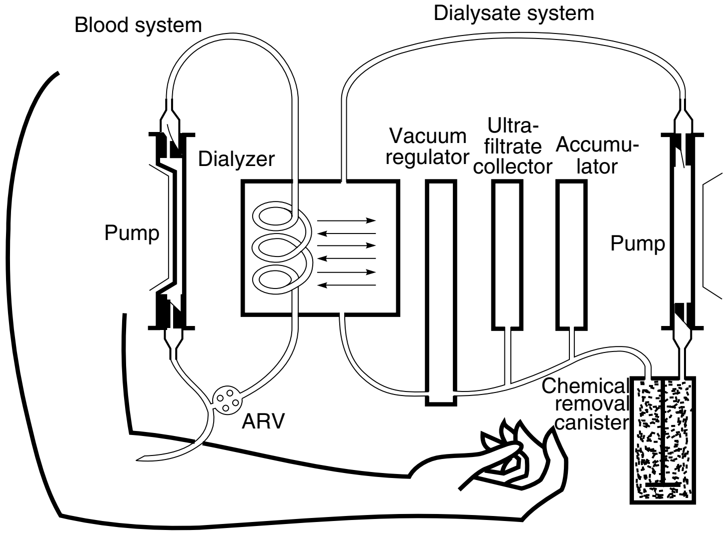
30.6 HOME (PORTABLE) KIDNEY MACHINES

There are two basic classes of dialysis machines: clinical units, which are commonly cabinet-size machines operated by trained technicians; and home-use dialysis machines, which are smaller and sometimes portable.

Normally, patients with complete loss of kidney function would need to visit the clinic at least three times per week and spend about four hours connected to the machine. With home-use machines, patients have more flexibility in scheduling dialysis, and they can dialyze for longer periods and more frequently. Thus, home-use machines are growing in popularity because they offer greater convenience and better clinical outcomes.

Over the years there have been efforts to develop machines for unsupervised home dialysis. At the same time, portability of the machine has been another consideration in the design of machines. The bulk of the machines is due to the dialysate preparation system and the electronic monitoring and control unit. The dialyzers have been reduced in size to a considerable extent. Kolff *et al.* (1976) report the development of a wearable artificial kidney (WAK) which uses a 20 l dialysate tank and 250 g of activated charcoal for a dialysate regenerating system. This has been possible with the development of battery-operated small-size pumps. The dialyzer used with the machine is hollow fibre dialyzer (1.4 m^2dialysis surface). The venous line bubble-catcher is attached to the outside of the dialyzer. In Fig. 30.16 the blood circuit is on the left and the dialyzing fluid circuit to the right. The dialysate is drawn from the dialyzer into the pump canister, past an accumulator which is a reservoir to expand and contract with the pulsation of the pump, then to an ultra-filtration valve which creates a negative pressure on the onflow line to the dialyzer and finally to the inflow of the dialyzer. The accumulator and ultra-filtrate reservoir do not function when the tank is in the circuit but are essential for use without the tank. Blood flow through the machine is simple, coming from the patient into the blood pump ventricle, to the inflow of the dialyzer, out of the dialyzer to the bubble-catcher and returning to the patient.

Modern home dialysis machines are designed to be more compact and easier to use than the large machines used in the clinics. As the purity of the water in the dialysis fluid is important, a significant feature of the machines is that they purify water from the tap without the need for significant modifications to plumbing. The dialysis fluid is prepared in batches, prior to dialysis. Ease of use in terms of cleaning, disinfecting, setting up and "stripping down" is achieved through the provision of a disposable cartridge which is inserted into the machine and discarded after each dialysis session. Transportability of these machines is made possible by



➤ Fig. 30.16 ***Wearable artificial kidney flow diagram (After Kolff et al., 1976)***

the availability of packs of pre-made dialysis fluid; the ability of the machines to use standard household electrical and plumbing systems; and the relatively small size of the machines approximately the size of a large suitcase.

A home dialysis machine generally consists of two parts: a portable dialysis machine or "cycler" through which the blood and dialysis fluid circulates and a device which purifies tap water and produces dialysis fluid. The device has flow rate of blood up to 600 ml/min, fluid exchange up to 12 L/hr and fluid removal up to 2.4 L/hr. Home-use dialysis machines need to include batteries to supplement the power supply's output power.

Researchers are exploring whether shorter daily session, or longer sessions performed overnight while the patient sleeps, are more effective in removing wastes. Newer dialysis machines make these alternatives more practical with home dialysis.

MODEL QUESTIONS

1. Explain the function of the kidneys. What changes take place in the body fluids in case of a renal disease?
2. What is the principle of dialysis in the artificial kidney? What are different types of dialyzers? Explain their construction and principle of operation.
3. What is the function of a membrane in the dialysis process? What are the different types of membranes used for dialysis?
4. What are the different performance parameters of dialyzers? Explain in detail.
5. Draw the block diagram of a haemodialysis machine and explain the importance of each building block.
6. What measurements are essential during the process of haemodialysis? Explain how these are carried out with the help of a diagram.
7. Distinguish between haemodialysis and peritoneal dialysis.